Fundamental Quantum Mechanics for Beginners

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Abstract

Quantum mechanics has revolutionized our understanding of the microscopic world and has become indispensable in modern engineering and technology—from semiconductor electronics and nanotechnology to quantum computing and quantum information. Recognizing the practical importance of mastering these concepts, this paper provides an accessible introduction to quantum mechanics for readers with a solid mathematical background. We begin by reviewing the concept of the wave function and the formulation of the Schrödinger Equation, which serve as the cornerstone for describing quantum systems. Next, we delve into the theory of observables and operators, exploring key concepts such as expectation values, variances, commutation relations, and the uncertainty principle. Finally, we illustrate these abstract ideas through classical models—including the infinite square well, the simple harmonic oscillator, and multi-dimensional systems—demonstrating how quantum mechanics not only explains long-standing puzzles but also forms the foundation for modern technological advances. We will also go over perturbation theorem which is one of the most useful theorem in not only theoretical but also application aspect.

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1 Introduction

Since Isaac Newton proposed his *Principia*, Newtonian mechanics has been considered one of the most systematic frameworks in physics. Although, later, Albert Einstein's theory of relativity showed that Newtonian mechanics fails at very high speeds or in strong gravitational fields and extended physics into a more general framework, the most significant idea—determinism—of the physical world still remains, meaning that given the initial conditions, one can always provide a precise prediction of a system's evolution.

However, as more experimental evidence emerged, scientists began to realize that classical mechanics cannot fully describe the world. For example, electron diffraction experiments revealed that electrons, which were expected to follow well-defined, classical trajectories, instead produced interference patterns when they hit a screen. Likewise, the *photoelectric effect* demonstrated that energy is emitted and absorbed in discrete packets—quanta—rather than continuously. Moreover, classical models of atomic structure predicted that accelerating electrons would continuously lose energy and ultimately spiral into the nucleus, contradicting the observed stability of atoms. These puzzling observations led researchers to gradually piece together new ideas that eventually culminated in the development of quantum mechanics.

One of the first breakthroughs came from *Max Planck*. To explain the phenomenon of *black body radiation*, he postulated that the energy of oscillators (atoms within molecules) could only take on discrete values rather than a continuous range. This idea of energy quantization formed the foundation of quantum theory and earned Planck the Nobel Prize in Physics in 1918. Later, in 1905, Albert Einstein extended Planck's theory by proposing the photon theory of the photoelectric effect. In his groundbreaking work, Einstein argued that light is composed of discrete packets of energy—photons—each carrying energy $E = hf$. This concept provided compelling evidence for energy quantization and earned him the Nobel Prize in Physics in 1921.

Ultimately, the gradual accumulation of such experimental anomalies and theoretical insights laid the foundation for quantum mechanics. One of the most important breakthroughs in this new framework was the derivation of the Schrödinger Equation by *Erwin Schrödinger* in 1926, which provides a fundamental description of the time evolution of a quantum system via its wave function—serving as the quantum analogue of Newton's Second Law. Together with the contributions of scientists such as *Dirac* and *Heisenberg*, quantum mechanics started to become a systematic science, thereby laying the foundation for modern physics.

Given these revolutionary breakthroughs, quantum mechanics has not only reshaped our understanding of the microscopic world but has also become indispensable in modern engineering and technology. Today, quantum theory underpins advances in fields ranging from semiconductor electronics and nanotechnology to quantum computing and communication. Recognizing the practical importance of mastering these concepts, I have designed this paper as an introduction for beginners with a solid mathematical foundation.

In the following sections, we will first introduce the wave function and the Schrödinger Equation to provide a foundational understanding of how quantum theory describes the world. Next, we delve into observables and operators to offer a more systematic and deeper insight into the quantum realm. Finally, we explore some classical models, which illustrate how

quantum mechanics is applied and how it explains various phenomena in nature.

2 State and Schrödinger's Equation

This section lays the foundation for understanding the quantum state and its evolution. We begin by introducing the wave function, the central concept in quantum mechanics that encapsulates the probabilistic nature of particles. Using Dirac notation, we describe how quantum states can be represented in both position and momentum spaces. We then illustrate how wave functions can be constructed from sinusoidal (trigonometric) bases, emphasizing their orthonormal properties, which allow for the unique decomposition of any periodic function. Furthermore, we discuss the Fourier transform as a powerful tool for connecting the position and momentum representations. Finally, the section introduces the Schrödinger equation, a fundamental postulate that replaces classical deterministic laws with a framework for quantum dynamics, forming the basis for the analysis of observables and operators in later sections.

2.1 Wave Functions

Quantum mechanics is fundamentally built upon the concept of the wave function, which encapsulates the inherently probabilistic nature of quantum states. Unlike classical mechanics—where a particle's state is precisely determined by its position and momentum—the quantum state is represented by a wave function that provides the probability amplitude for finding a particle at a given location or with a given momentum. In this section, we introduce the wave function and its concise representation using Dirac notation, then describe how wave functions can be constructed using sinusoidal functions, highlighting the key property of orthogonality and the role of orthonormal bases in decomposing periodic functions. Finally, we discuss the Fourier transform, which not only bridges the position and momentum representations but also underscores the underlying linear algebraic structure of quantum mechanics.

2.1.1 Wave Functions and its Dirac Notation

In classical mechanics, the state of a particle can be completely determined by specifying its position and momentum at a given time, allowing one to predict the future behavior of the system precisely. In contrast, quantum mechanics introduces the concept of the *wave function* to describe the state of a particle. However, the behavior of wave functions is fundamentally different from the deterministic description in classical mechanics. In quantum mechanics, a state can be represented independently by either its position wave function or its momentum wave function:

$$\psi(x,t), \quad \psi(p,t), \tag{2.1.1.1}$$

both of which are complex-valued functions that encapsulate all the information about the quantum system. We refer to these as the *position representation* and the *momentum representation*.

Taking the position representation as an example, the wave function $\psi(x,t)$ does not specify an exact location; rather, it provides the probability amplitude for finding the particle at position x at time t. According to the *Born rule*, the probability density for finding the particle at x is given by

$$\rho(x,t) = |\psi(x,t)|^2. \tag{2.1.1.2}$$

Thus, the probability of finding the particle within the interval $[a,b]$ is

$$P(a \le x \le b) = \int_a^b |\psi(x,t)|^2 \, dx. \tag{2.1.1.3}$$

In addition to representing a state through $\psi(x,t)$ or $\psi(p,t)$, quantum mechanics employs *Dirac notation*. In this formalism, the wave function is expressed as a column vector $|\psi\rangle$, called a *ket* vector. Since every wave function is complex-valued, it has a conjugate transpose, denoted by $\langle\psi|$, called a *bra* vector. We denote the conjugate transpose as

$$|\psi\rangle^\dagger = \langle\psi| \,. \tag{2.1.1.4}$$

2.1.2 Sinusoidal Basis and Orthonormality

The term *wave function* naturally suggests that the state of a quantum system behaves like a wave. In many physical systems, the solutions to the Schrödinger equation are oscillatory, resembling classical waves. The sine and cosine functions, with their periodic behavior, serve as the fundamental building blocks in Fourier analysis and are often used to construct wave functions—especially in systems with periodic or confined spatial domains.

A key reason for their usefulness is that sine and cosine functions satisfy an important mathematical property known as *orthogonality*. In linear algebra, a set of functions $\{f_n(x)\}$ is said to be orthogonal over an interval $[a,b]$ if

$$\int_a^b f_m(x) f_n(x) \, dx = 0 \quad \text{for } m \neq n. \tag{2.1.2.1}$$

If each function is also normalized such that

$$\int_a^b f_n(x)^2 \, dx = 1, \tag{2.1.2.2}$$

then the set is called *orthonormal*. The process of adjusting a wave function so that its total probability is unity is known as *Normalization*. Specifically, for a wave function $\psi(x)$ defined over an interval $[a,b]$, normalization requires that

$$\int_a^b |\psi(x)|^2 \, dx = 1, \tag{2.1.2.3}$$

and if this condition is not met, one obtains the normalized wave function $\psi_{\text{norm}}(x)$ by

$$\psi_{\text{norm}}(x) = \frac{\psi(x)}{\sqrt{\int_a^b |\psi(x)|^2 \, dx}}. \tag{2.1.2.4}$$

The concept of orthonormality is crucial because it allows us to represent a general wave function as a unique linear combination of these basis functions, much like expressing any vector in $\mathbb{R}^n$ using an orthonormal basis. In fact, any periodic function can be represented as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \tag{2.1.2.5}$$

(For a detailed proof of the orthogonality of trigonometric functions, see Appendix A.)

Since any periodic function can be represented by (2.1.2.5), we determine the coefficient of each component by projecting the function onto the direction of each basis function. In

general, for a function y represented by an orthogonal basis $\{v_i\}$, we have

$$y = \sum_{i=1}^{k} \frac{\langle y, v_i \rangle}{\|v_i\|^2} v_i. \tag{2.1.2.6}$$

In the trigonometric case, this leads to the Euler-Fourier formulas for the coefficients:

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\, dx, \\ a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} \cos(nx) f(x)\, dx, \\ b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} \sin(nx) f(x)\, dx. \end{aligned} \tag{2.1.2.7}$$

Furthermore, in quantum mechanics another convenient representation is the exponential form, derived from Euler's formula. In this representation,

$$\begin{aligned} f(x) &= \sum_{n=-\infty}^{\infty} C_n e^{in\pi x/L}, \\ C_n &= \frac{1}{2L} \int_{-L}^{L} e^{-in\pi x/L} f(x)\, dx, \end{aligned} \tag{2.1.2.8}$$

where the function has a period of $2L$ (i.e., over the interval $[-L, L]$) instead of 2π (i.e., $[-\pi, \pi]$). (For a detailed proof of the exponential representation, see Appendix B.)

2.1.3 Fourier Transform and Momentum Representation

Under the exponential representation, a plane wave with wave number

$$k = n\frac{\pi}{L}$$

is given by

$$e^{ikx}. \tag{2.1.3.1}$$

The corresponding momentum is

$$p_n = \frac{h}{\lambda} = \hbar k_n = \frac{\hbar \pi n}{L}. \tag{2.1.3.2}$$

Thus, the expression

$$e^{ip_n x/\hbar} \tag{2.1.3.3}$$

represents a plane wave with position x and momentum p_n. Together with the continuous form of the Fourier transform (see Equation (2.1.2.8)), we can establish the relationship between the position and momentum representations:

$$\psi(x,t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} \psi(p,t) e^{ipx/\hbar}\, dp, \tag{2.1.3.4}$$

$$\psi(p,t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} \psi(x,t) e^{-ipx/\hbar}\, dx. \tag{2.1.3.5}$$

This pair of equations defines the Fourier transform and its inverse. The Fourier transform is an essential tool in connecting the position and momentum representations, as it allows us to express the wave function in the position representation, $\psi(x,t)$, as a linear combination of momentum eigenstates [see Equation (2.1.3.4)], and conversely, to obtain the momentum wave function $\psi(p,t)$ from $\psi(x,t)$ via Equation (2.1.3.5).
Moreover, the mathematical structure of the Fourier transform underpins the uncertainty relation between position and momentum, a topic that will be discussed in more detail in later sections. (For a detailed proof of the Fourier transform relationship, see Appendix C.)

2.2 Schrödinger Equation and Eigenvalue Problems

In this subsection, we introduce the fundamental role of the Schrödinger equation in quantum mechanics. Similar to how Newton's Second Law is postulated as a basic principle in classical mechanics—despite appearing to arise "ex nihilo"—the Schrödinger equation is introduced as an axiomatic postulate that governs the evolution of quantum states. We begin by discussing the motivation for replacing the deterministic Newtonian framework with a probabilistic description based on wave functions. Then, we present the general (time-dependent) Schrödinger equation and explain its significance as a cornerstone of quantum theory. Finally, we focus on stationary states and the method of separation of variables, which lead to an eigenvalue problem whose solutions determine the quantized energy levels of the system. This structured approach not only highlights the conceptual parallels between classical and quantum mechanics but also lays the groundwork for further discussions on observables and operators in subsequent sections.

2.2.1 Introduction of Schrödinger's Equation and Fundamental Ideas

In classical mechanics, the fundamental equation is Newton's Second Law,

$$F = ma = m\frac{dp}{dt}, \tag{2.2.1.1}$$

which appears as an empirical postulate without derivation from first principles. Over time, extensive experiments and theoretical work have confirmed its validity, establishing it as a cornerstone of classical mechanics.
However, when we represent the state of a particle or a system using a wave function, the concept of a well-defined trajectory and force is no longer applicable. In this context, Newton's Second Law becomes inadequate, and a new fundamental postulate is required.
In the previous section, we saw that once the wave function ψ of a system is determined, all physical quantities—including the probability distribution and average values (which we will explore in later sections)—are fixed. Therefore, the key task in quantum mechanics is to determine all possible wave functions that describe the quantum state and to understand how $\psi(x,t)$ evolves over time.
The *Schrödinger equation*, proposed by Schrödinger in 1926, emerged as an axiomatic postulate—much like Newton's Second Law—and successfully addresses this need:

$$i\hbar \frac{\partial \Psi(\mathbf{x},t)}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2 \Psi(\mathbf{x},t) + V(\mathbf{x})\Psi(\mathbf{x},t). \tag{2.2.1.2}$$

It is important to note that both Newton's Second Law and the Schrödinger equation were introduced as fundamental postulates, seemingly arising "ex nihilo." Their ultimate justification came through experimental validation and internal consistency within their respective theoretical frameworks. Just as mathematics is built upon a set of axioms, these equations form the foundational assumptions of classical and quantum mechanics, respectively.

2.2.2 Stationary State

In the general Schrödinger equation, the potential V may depend on time. However, for a large class of problems—the stationary states—the potential is time-independent, i.e., $V = V(\mathbf{x})$.

In such cases, we can seek solutions by separation of variables. Assume a solution of the form

$$\Psi(\mathbf{x},t) = \psi(\mathbf{x})f(t). \tag{2.2.2.1}$$

Substituting into the time-dependent Schrödinger equation, one obtains two separate equations. The time-dependent part yields

$$f(t) = e^{-iEt/\hbar}, \tag{2.2.2.2}$$

where E is a constant representing the energy eigenvalue. The spatial part then satisfies the stationary (time-independent) Schrödinger equation:

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{x}) + V(\mathbf{x})\psi(\mathbf{x}) = E\psi(\mathbf{x}). \tag{2.2.2.3}$$

This eigenvalue problem determines the stationary states of the system along with their corresponding energy levels E. The eigenfunctions $\psi_E(x)$ form a complete orthonormal set, allowing any quantum state to be expressed as a linear combination of these energy eigenstates:

$$\Psi(\mathbf{x},t) = \sum_E C_E\psi_E(\mathbf{x})e^{-iEt/\hbar}. \tag{2.2.2.4}$$

(For a detailed derivation of the stationary state solution, see Appendix D.)

2.2.3 Conservation of Probability

As Schrödinger's equation described the change of wave function over space, since wave function can give us the probability of a particle appear in the space, it is crucial to learn about how probability changed over space.
As the probability density of finding a particle in $\mathbf{x}$ at time t is given by

$$\rho(\mathbf{x},t) = \psi^*(\mathbf{x},t)\psi(\mathbf{x},t) \tag{2.2.3.1}$$

and as we want to learn about how this probability density changed with time, we do the partial derivative with respect to t

$$\frac{\partial}{\partial t}\rho(\mathbf{x},t) = \frac{\partial\psi^*(\mathbf{x},t)}{\partial t}\psi(\mathbf{x},t) + \frac{\partial\psi(\mathbf{x},t)}{\partial t}\psi^*(\mathbf{x},t) \tag{2.2.3.2}$$

According to Schrödinger's equation, we obtain

$$\begin{aligned}\frac{\partial}{\partial t}\psi(\mathbf{x},t) &= \frac{i\hbar}{2m}\nabla^2\psi + \frac{1}{i\hbar}V\psi\\ \frac{\partial}{\partial t}\psi^*(\mathbf{x},t) &= \frac{-i\hbar}{2m}\nabla^2\psi - \frac{1}{i\hbar}V\psi\end{aligned} \tag{2.2.3.3}$$

Substitute back into (2.2.3.2),

$$\begin{aligned}\frac{\partial}{\partial t}\rho(\mathbf{x},t) =& \frac{i\hbar}{2m}[\psi^*\nabla^2\psi - (\nabla^2\psi^*)\psi] + \frac{1}{i\hbar}(\psi^*V\psi - \psi^*V^*\psi)\\ =& \frac{i\hbar}{2m}\nabla\cdot[\psi^*\nabla\psi - (\nabla\psi^*)\psi] + \frac{1}{i\hbar}(V-V^*)\psi^*\psi\\ =& -\nabla\cdot\mathbf{J} + \frac{1}{i\hbar}(V-V^*)\rho\end{aligned} \tag{2.2.3.4}$$

Where we have

$$\mathbf{J} = \frac{i\hbar}{2m}[(\nabla\psi^*)\psi - \psi^*\nabla\psi] \tag{2.2.3.5}$$

And since generally, we only consider real valued potential function, thus we have $V = V^*$ Thus, we obtain

$$\frac{\partial}{\partial t}\rho + \nabla\cdot\mathbf{J} = 0 \tag{2.2.3.6}$$

And one can also have a integral form

$$\frac{d}{dt}\iiint_V \rho\, d\tau = -\oiint_S \mathbf{J}\cdot d\mathbf{S} \tag{2.2.3.7}$$

Known as continuity equation, if you've been studied electrodynamic, you will be familiar with this form of equation, in electrodynamic, the continuity equation describes the conservation of charges; similarly, in our case, this continuity equation tells us the conservation of probability. It is important to notice that, this conservation of probability requires the real valued potential function; when discussing the absorption in atomic reaction, it can have complex valued potential functions, in that case, the probability is no longer conserved.

3 Observables and Operators

In the previous sections, we introduced the state function of quantum system, which can be used to fully describe a quantum system, and Schrödinger equation can be used to describe how quantum state evolved over time. However, in real life application, it is usually hard to obtain the exact wave function of a system, when doing experiment, the only thing we can obtain by measurement are some observables. Then, in order to figure out, how to find these observables when knowing the wave functions; when we have observables, how to determine the properties of the wave function, and when a wave function is representing the distribution of different observables, what's the relationship between them, we are going to delve deep into the observables and operators.

3.1 The Average of Observables

In classical mechanics, the analysis of a system is often simplified by considering its center of mass rather than tracking every detail of its motion. Similarly, in quantum mechanics, since the state of a system is represented by a wave function that describes a probability distribution, it is often useful to consider the average (or expectation) values of observables such as position or momentum.

3.1.1 Average of Position

According to classical mechanics, the average position of particles is

$$\bar{x} = \langle x\rangle = \frac{\int xN(x)\,dx}{\int N(x)\,dx} \tag{3.1.1.1}$$

Where $N(x)$ is representing the number of particle on position x. As we introduce the statistical nature of wave function, we have the total number of particle under volume differential $d\tau = dxdydz$

$$N(x)d\tau = |\psi(\mathbf{x},t)|^2 d\tau = \psi^*(\mathbf{x},t)\psi(\mathbf{x},t)dxdydz \tag{3.1.1.2}$$

Substitute this relation into (3.1.1.1)

$$\begin{aligned}\bar{x} = \langle x\rangle =& \frac{\iiint x\psi^*(\mathbf{x},t)\psi(\mathbf{x},t)\,dxdydz}{\iiint \psi^*(\mathbf{x},t)\psi(\mathbf{x},t)\,dxdydz}\\ =& \iiint \frac{\psi^*(\mathbf{x},t)}{\sqrt{N}}x\frac{\psi(\mathbf{x},t)}{\sqrt{N}}\,d\tau\end{aligned} \tag{3.1.1.3}$$

If our wave function are already normalized, we have $N = 1$, then

$$\bar{x} = \iiint \psi^*(\mathbf{x},t)\,x\,\psi(\mathbf{x},t)\,d\tau \tag{3.1.1.4}$$

And similar with other two coordinate

$$\begin{aligned}\bar{y} &= \iiint \psi^*(\mathbf{x},t)\, y\, \psi(\mathbf{x},t)\, d\tau \\ \bar{z} &= \iiint \psi^*(\mathbf{x},t)\, z\, \psi(\mathbf{x},t)\, d\tau\end{aligned} \tag{3.1.1.5}$$

This strategy can also used to find the average of a measurable physical quantity

$$\bar{F}(t) = \iiint \psi^*(\mathbf{x},t)\, F(\mathbf{x})\, \psi(\mathbf{x},t)\, d\tau \tag{3.1.1.6}$$

3.1.2 Average of Momentum

The way of finding average of momentum is a little different with what you might think. But technically, it is similar with what we did to find the average of position.
From the (2.1.3.5) we have the Fourier components of $\psi(x,t)$. Then the probability for a particle to have momentum p is

$$w(p) = |C(p,t)|^2 \tag{3.1.2.1}$$

With the idea inherited from before

$$\begin{aligned}\bar{p} = \langle p\rangle &= \int p|C(p,t)|^2\, dp \\ &= \frac{1}{2\pi\hbar}\iiint \psi^*(x',t)\, e^{ipx'/\hbar}\, p\, \psi(x,t)\, e^{-ipx/\hbar} dx' dx dp\end{aligned} \tag{3.1.2.2}$$

And since

$$pe^{ipx/\hbar} = i\hbar\frac{d}{dx}(e^{-ipx/\hbar})dx \tag{3.1.2.3}$$

$$\begin{aligned}\int_{-\infty}^{\infty}\psi(x,t)pe^{ipx/\hbar}\, dx &= \int_{-\infty}^{\infty}\psi(x,t)i\hbar\frac{d}{dx}(e^{-ipx/\hbar})\, dx \\ &= i\hbar\psi(x,t)e^{-ipx/\hbar}\Big|_{-\infty}^{\infty} - \int e^{ipx/\hbar}i\hbar\frac{d}{dx}\psi(x,t)\, dx\end{aligned} \tag{3.1.2.4}$$

Since the integral is over the space where particle exists, thus, out of the space of the integral, the wave function will be zero, thus the first item diminished. Substitute into (3.1.2.2)

$$\begin{aligned}\bar{p} &= \frac{1}{2\pi\hbar}\int \psi^*(x',t)e^{-ip(x'-x)/\hbar}\, dx' dp(-i\hbar)\frac{d}{dx}\psi(x,t)dx \\ &= \int \psi^*(x,t)\frac{\hbar}{i}\frac{d\psi(x,t)}{dx}dx\end{aligned} \tag{3.1.2.5}$$

In three dimensional case, we have

$$\langle\mathbf{p}\rangle = \int \psi^*(\mathbf{x},t)\frac{\hbar}{i}\nabla\psi(\mathbf{x},t)\, d\tau \tag{3.1.2.6}$$

Thus, now, after our discussion in this section, we obtained and define two basic operator of quantum mechanics

$$\begin{cases}\hat{x}_i = x \\ \hat{p}_i = \frac{\hbar}{i}\frac{\partial}{\partial x_i}\end{cases} \tag{3.1.2.7}$$

3.2 Introduction to Operators

3.2.1 Basic Properties of Operators

In the previous section, we introduced *operators* when discussing how to compute the average values of position and momentum. In quantum mechanics, if the state of a system changes due to some operation, the physical quantity that represents this change is described by an operator. In general, if Ψ is the state of the system, an operation $\hat{A}$ acting on this state produces a new state Φ:

$$\Phi = \hat{A}\Psi. \tag{3.2.1.1}$$

As we have learned in linear algebra, all the processes in quantum mechanics occur in a Hilbert space (which we will discuss in a later section), and the operations are inherently linear. Moreover, many of these operations are unitary, ensuring that probabilities are conserved. Beginning from this section, we will delve deeply into the properties and applications of operators in quantum mechanics. If you are familiar with linear algebra, you will notice many familiar concepts and techniques reappear in the quantum context.

Equality between Operators
If operator $\hat{A}$ and $\hat{B}$ perform on any state ψ end up with same state, then we say $\hat{A}, \hat{B}$ are equal.

$$\forall\,\psi,\ \hat{A}\psi = \hat{B}\psi \Rightarrow \hat{A} = \hat{B} \tag{3.2.1.2}$$

Addition of Operators

$$\forall\,\psi,\ \hat{C}\psi = \hat{A}\psi + \hat{B}\psi \Rightarrow \hat{C} = \hat{A} + \hat{B} \tag{3.2.1.3}$$

Multiplication of Operators

$$\forall\,\psi,\ \hat{A}(\hat{B}\psi) = \hat{C}\psi \Rightarrow \hat{A}\hat{B} = \hat{C} \tag{3.2.1.4}$$

Linear Operators
If ψ_1, ψ_2 are wave functions describing two state, then, their linear combination, since superposition principle, is also a state.

$$\psi = C_1\psi_1 + C_2\psi_2 \tag{3.2.1.5}$$

where C_1, C_2 are any complex number.
And if an Operator $\hat{A}$ can keep this condition satisfies, it can be called a linear operator.

$$\hat{A}\psi = C_1\hat{A}\psi_1 + C_2\hat{A}\psi \tag{3.2.1.6}$$

3.2.2 Hermitian Operator

If you have studied linear algebra, you may be familiar with a type of operator called *Hermitian* (or *self-adjoint*). An operator $\hat{A}$ is Hermitian if it satisfies

$$\hat{A}^\dagger = \hat{A}, \tag{3.2.2.1}$$

which in terms of integrals can be expressed as

$$\int \psi^*\,\hat{A}^\dagger\phi\, d\tau = \int (\hat{A}\psi)^*\,\phi\, d\tau, \tag{3.2.2.2}$$

for all appropriate functions ψ and ϕ. A significant example is the position operator $\hat{x}$, which is clearly Hermitian. Consider also the momentum operator $\hat{p}$, defined by

$$\hat{p} = \frac{\hbar}{i}\frac{d}{dx}.$$

Its Hermiticity can be verified as follows:

$$\begin{aligned}\int \psi^*\hat{p}\,\phi\, dx &= \int \psi^*\left(\frac{\hbar}{i}\frac{d\phi}{dx}\right)dx \\ &= \frac{\hbar}{i}\,\psi^*\phi\Big|_{-\infty}^{+\infty} - \int \frac{\hbar}{i}\frac{d\psi^*}{dx}\,\phi\, dx \\ &= \int \left(\frac{\hbar}{i}\frac{d\psi}{dx}\right)^*\phi\, dx,\end{aligned} \tag{3.2.2.3}$$

assuming that the boundary term vanishes. This shows that $\hat{p}$ is Hermitian.

Properties of Hermitian Operators:

1. **Reciprocity:** If $\hat{A}^\dagger = \hat{B}$, then $\hat{A} = \hat{B}^\dagger$.
2. **Reversed Order:** For a product of operators,
$$(\hat{A}\hat{B}\hat{C}\cdots)^\dagger = \cdots \hat{C}^\dagger \hat{B}^\dagger \hat{A}^\dagger. \tag{3.2.2.4}$$
3. **Reversed Order on Commutator:**
$$[\hat{A}, \hat{B}]^\dagger = [\hat{B}^\dagger, \hat{A}^\dagger] = -[\hat{A}^\dagger, \hat{B}^\dagger]. \tag{3.2.2.5}$$
4. **Real-Valued Eigenvalues:** Every eigenvalue of a Hermitian operator is real.
5. **Orthogonality of Eigenstates:** Eigenstates corresponding to different eigenvalues of a Hermitian operator are orthogonal.
6. **Orthonormal Basis for Same Eigenvalue:** The linearly independent eigenstates corresponding to the same eigenvalue can be chosen to be orthonormal.
7. **Ordering of Eigenvalues:** The eigenvalues of a Hermitian operator can be arranged in increasing order.

(The proof of 4-6 can be found in Appendix F)

3.3 Variance, Eigenstate, Eigenvalue

3.3.1 Variance

Although the average (expectation) value can reflect some basic properties of a system, many physical quantities may have very different distributions even when they share the same average value (see Figure 1). In such cases, the average alone is insufficient to fully characterize the system. Therefore, we introduce a statistical measure called *variance*, which quantifies the spread of a distribution.

The idea is straightforward: for a given physical quantity F, we first define the deviation from the average as

$$\Delta F = F - \bar{F}, \tag{3.3.1.1}$$

where $\bar{F}$ is the average value of F. To characterize the overall spread of F in the system, we consider either the average of the absolute deviations,

$$\langle |\Delta F| \rangle, \tag{3.3.1.2}$$

or more commonly, the average of the squared deviations:

$$\langle (\Delta F)^2 \rangle = \overline{\Delta F^2}. \tag{3.3.1.3}$$

This quantity, $\overline{\Delta F^2}$, is known as the *variance*.

In the context of quantum mechanics, the variance of an observable F is defined as

$$\overline{\Delta F^2} = \int \psi^* (\hat{F} - \overline{F})^2 \psi \, d\tau, \tag{3.3.1.4}$$

where ψ is the wave function of the system, $\hat{F}$ is the operator corresponding to the observable F, $\overline{F}$ is the expectation value of F, and $d\tau$ denotes the integration over the relevant configuration space.

The variance provides a measure of the distribution's width: a larger variance indicates a wider spread, meaning that the values of F are more dispersed around the mean.

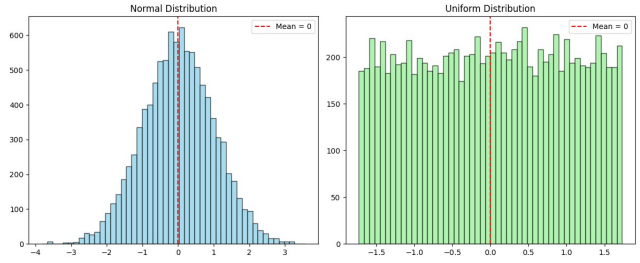


Figure 1: Same average value, but different distribution

3.3.2 Eigen-Wave-Function and Eigenvalue

Generally speaking, when an operator acts on a quantum state, the state is transformed into another state. However, there exist special states that remain unchanged by the action of a given operator, except for a scaling factor. In linear algebra, such states are known as eigenvectors, and the corresponding scaling factors are called eigenvalues. In quantum mechanics, these states are represented by wave functions, leading to the eigenvalue equation

$$\hat{A}\psi = \lambda\psi, \tag{3.3.2.1}$$

where λ is a number (the eigenvalue) and ψ is called an *eigen-wave-function*. This invariance (up to a scalar multiple) reflects intrinsic properties of the observable associated with the operator $\hat{A}$. Consequently, we refer to a state that remains unchanged (except for scaling) under $\hat{A}$ as an *eigenstate* of the observable, with λ being its corresponding eigenvalue.

3.3.3 Eigen-Spectrum and Degeneracy

The complete set of eigenvalues of an operator is called its *eigen-spectrum* or simply its *spectrum*. Depending on the operator, the spectrum may be discrete or continuous. When the eigenvalues are discrete, they form a *discrete spectrum*; when they are continuous, they form a *continuous spectrum*. For a given eigenvalue λ, the operator may have one or more eigen-wave-functions. If there exist f linearly independent eigen-wave-functions, say $\varphi_1, \varphi_2, \ldots, \varphi_f$, associated with the same eigenvalue λ, we say that λ has a degeneracy of f (or is f-fold degenerate). In other words, the existence of multiple independent eigenstates corresponding to the same eigenvalue is referred to as *degeneracy*.

3.3.4 Eigenstate

Consider a quantum state in which every measurement of a particular observable yields the same value F. In such a state, the expectation value of the observable is exactly F, and the uncertainty (variance) is zero:

$$\overline{F} = F, \quad \overline{\Delta F^2} = 0. \tag{3.3.4.1}$$

What does this imply about the behavior of the wave function?

Since the operators corresponding to observables are Hermitian, we have

$$\begin{aligned} (\Delta\hat{F})^\dagger &= (\hat{F} - \overline{F})^\dagger = \hat{F}^\dagger - \overline{F} = \Delta\hat{F}, \\ \langle \Delta F^2 \rangle &= \int \psi^* (\Delta\hat{F})^2 \psi \, d\tau = \int \psi^* (\Delta\hat{F})(\Delta\hat{F})\psi \, d\tau \\ &= \int (\Delta\hat{F}\psi)^* (\Delta\hat{F}\psi) \, d\tau = \int |\Delta\hat{F}\psi|^2 \, d\tau. \end{aligned} \tag{3.3.4.2}$$

Since $|\Delta\hat{F}\psi|^2$ is a non-negative quantity, for the variance to vanish we must have

$$\Delta\hat{F}\psi = 0, \tag{3.3.4.3}$$

or equivalently,

$$(\hat{F} - \overline{F})\psi = 0, \tag{3.3.4.4}$$

which can be rewritten as

$$\hat{F}\psi = \overline{F}\,\psi. \tag{3.3.4.5}$$

This is precisely the eigenvalue equation for the observable F, where the average $\overline{F}$ represents the definite value of F obtained in every measurement, and ψ is the eigen-wavefunction corresponding to this eigenvalue.
Thus, we can conclude that a sufficient and necessary condition for an observable to have a well-defined, confirmed value is that the system is in an eigenstate of the corresponding operator. In this case, the confirmed value is exactly the eigenvalue associated with that eigenstate.

3.3.5 Linearity of Operator Representing a Observable

In the last section, we said that the operator representing an observable should be Hermitian. This is because the eigenvalue of this operator should be the value of these observables, which should be real-valued. And since the property of the Hermitian operator, the Hermitian operator also has real-valued eigenvalue, which coincides with what is required for an observable operator. Furthermore, we can also prove that, to align with the superposition principle of state, the operator representing a observable should also be linear.
Suppose there are two state ψ_1, ψ_2, representing a observable with confirmed value f in two different state

$$\begin{cases} \hat{F}\psi_1 = f\psi_1 \\ \hat{F}\psi_2 = f\psi_2 \end{cases} \tag{3.3.5.1}$$

With the superposition principle, we have

$$\psi = c_1\psi_1 + c_2\psi_2 \tag{3.3.5.2}$$

As we learned in linear algebra, since it is a linear combination within the eigenspace, it should still have same eigenvalue f.

$$\hat{F}\psi = f\psi \tag{3.3.5.3}$$

which should satisfy

$$\hat{F}\psi = \hat{F}(c_1\psi_1 + c_2\psi_2), \quad f\psi = fc_1\psi + fc_2\psi \tag{3.3.5.4}$$

$$f\psi = c_1\hat{F}\psi_1 + c_2\hat{F}\psi_2 \tag{3.3.5.5}$$

$$\hat{F}(c_1\psi_1 + c_2\psi_2) = c_1\hat{F}\psi_1 + c_2\hat{F}\psi_2 \tag{3.3.5.6}$$

Which we obtained the linearity operator defined in (3.2.1.6). Thus, we conclude that the operator representing a observable should be linear and Hermitian, which is also a part of requirement for the *Third Postulate of Quantum Mechanics*.

3.4 Commutative Relation, Anti-Commutative Relation and Uncertainty Principle

3.4.1 Commutative Relation

In linear algebra, there is a term called *commutation*, which refers to the property that two operators can be interchanged without affecting the outcome of their combined operation. Formally, for two operators $\hat{A}$ and $\hat{B}$, the commutator is defined as

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}.$$

If $[\hat{A}, \hat{B}] = 0$, then $\hat{A}$ and $\hat{B}$ are said to *commute*. In quantum mechanics, commuting operators typically correspond to observables that can be simultaneously measured with arbitrary precision, since they share a common set of eigenstates. Under this *commute* nature, the change of order of operation has no different on the result.
(For the proof of common eigenstate when commute, see Appendix E)
An example of this is $\hat{x}$ and $\hat{p}$

$$\hat{p}\hat{x}\psi = -i\hbar\frac{d}{dx}(x\psi) = -i\hbar\psi - i\hbar x\frac{d}{dx}\psi = -i\hbar\psi + \hat{x}\hat{p}\psi \tag{3.4.1.1}$$

Thus,

$$\hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar \tag{3.4.1.2}$$

Thus, a different order of apply this two operation would cause difference, which led to the *Uncertainty Principle* and we will discuss it in later section.

Properties of Commutators

1. Anti-Symmetric

$$[\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}] \tag{3.4.1.3}$$

2. Linearity

$$[\hat{A}, \alpha\hat{B} + \beta\hat{C}] = \alpha[\hat{A}, \hat{B}] + \beta[\hat{A}, \hat{C}] \tag{3.4.1.4}$$

3. Operation by Parts

$$[\hat{A}, \hat{B}\,\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}] \tag{3.4.1.5}$$

4. Jacobian Identity

$$[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0 \tag{3.4.1.6}$$

The commutator is not transitive, in many cases, it can be commute between $\hat{A}, \hat{B}$ and $\hat{B}, \hat{C}$, but $\hat{A}, \hat{C}$ not guaranteed to be commutable.

3.4.2 Anti-Commutative Relation

Conversely, there is the concept of *anti-commutation*, which describes a situation where the order of operations matters such that swapping the order results in a sign change. Two operators $\hat{A}$ and $\hat{B}$ are said to *anti-commute* if their anti-commutator vanishes:

$$\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A} = 0.$$

Anti-commutation relations play a crucial role in the study of fermionic systems in quantum mechanics and quantum field theory. For example, the creation and annihilation operators for fermions obey anti-commutation relations to enforce the Pauli exclusion principle, ensuring that no two identical fermions occupy the same quantum state.

Properties of Anti-Commutators

1. Symmetric

$$\{\hat{A}, \hat{B}\} = \{\hat{B}, \hat{A}\} \tag{3.4.2.1}$$

2. Linearity

$$\{\hat{A}, \alpha\hat{B} + \beta\hat{C}\} = \alpha\{\hat{A}, \hat{B}\} + \beta\{\hat{A}, \hat{C}\} \tag{3.4.2.2}$$

3. Operation by Parts

$$\{\hat{A}, \hat{B}\,\hat{C}\} = \{\hat{A}, \hat{B}\}\hat{C} - \hat{B}\{\hat{A}, \hat{C}\} \tag{3.4.2.3}$$

4. The product of any two operator is equal to the addition of commutator and anti-commutator

$$\hat{A}\hat{B} = \frac{1}{2}[\hat{A}, \hat{B}] + \frac{1}{2}\{\hat{A}, \hat{B}\} \tag{3.4.2.4}$$

3.4.3 Heisenberg Uncertainty Relation

In previous sections, we introduced commutator, if observable $\hat{F}, \hat{G}$ are commute

$$[\hat{F}, \hat{G}] = 0 \tag{3.4.3.1}$$

Which means that their variance is 0

$$(\overline{\Delta F})^2 = 0, \quad (\overline{\Delta G})^2 = 0 \tag{3.4.3.2}$$

This can be conclude as, we have the average value of this two operator equals to the value of each operator. We can also conclude that, under such condition, the distribution of two operator are centralized.

However, many fundamental observables in quantum mechanics are non-commuting. A significant example is the position operator $\hat{x}$ and the momentum operator $\hat{p}$, which we have

$$[\hat{x}, \hat{p}] = i\hbar. \tag{3.4.3.3}$$

This non-commutativity is at the core of the Heisenberg uncertainty principle. But where does this uncertainty originate? The answer lies in the wave-particle duality of quantum objects.

In quantum mechanics, particles such as electrons or photons exhibit both particle-like and wave-like behavior. When we attempt to localize a particle by confining its wave function to a narrow region in space—say, by passing it through a narrow slit—we inherently invoke its wave nature. A narrow slit acts as a spatial constraint, and according to the principles of wave diffraction, such confinement leads to a spread in the particle's momentum. In other words, the more precisely we localize the particle in position (reducing Δx), the more its momentum distribution must spread out (increasing Δp).

This effect can be understood in terms of the Fourier transform. The wave function in position space and its counterpart in momentum space are Fourier transform pairs. A function that is highly localized in position space (a narrow wave packet) necessarily has a broad Fourier transform, meaning that its momentum distribution is spread over a wide range of values.(Figure 2) Mathematically, this relationship is expressed by the uncertainty principle:

$$\Delta x\, \Delta p \geq \frac{\hbar}{2}. \tag{3.4.3.4}$$

Thus, the inability to simultaneously determine both the position and momentum with arbitrary precision is a direct consequence of the wave-like nature of particles and the properties of the Fourier transform.

In summary, the uncertainty principle reflects the fundamental wave-particle duality: confining a wave (particle) in space (i.e., reducing Δx) leads to an increased spread in its momentum distribution (Δp), as evidenced by phenomena such as diffraction. (For a detailed derivation of the uncertainty principle, see Appendix G.) This interplay between localization and momentum spread is at the heart of quantum mechanics, setting it apart from the deterministic picture of classical physics.

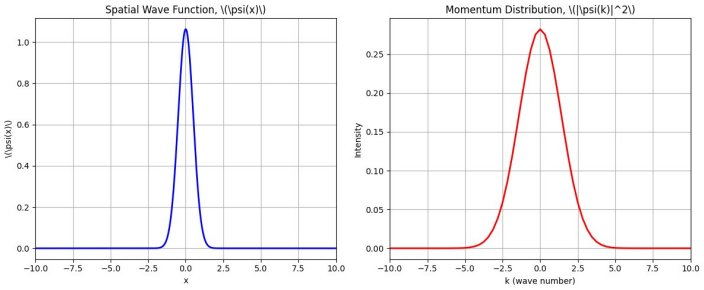


Figure 2: The relationship between spacial wave function(Left) and momentum space distribution(Right)

3.5 The Completeness of Eigenstate for Observable

We already state that the wave function is a sufficient description of a state of a quantum system, however, we still haven't answer the question on: how can we derive the distribution of a observable in a quantum system? In the following sections, we are going to answer this question.

3.5.1 Completeness of Eigenstate

In the electron diffraction experiment, due to the particle nature of electrons, each electron is observed on the screen at a definite position. This means that, upon measurement, the electron collapses into a position eigenstate with a well-defined coordinate. Importantly, any state in this coordinate system can be expressed as a superposition of a complete set of orthogonal position eigenstates. Thus, to determine the distribution of positions, we examine the magnitude of the projection onto each orthogonal component of the position basis.

Building on this idea, we extend the strategy to describe any observable. By expanding a quantum state in the eigenbasis of an observable, we leverage the orthogonality of the eigenstates, ensuring that each component is independent and does not interfere with the others. This orthogonal decomposition is crucial because it simplifies the analysis and prediction of measurement outcomes in quantum mechanics.

Let's begin on this topic by illustrate the general meaning of this expansion, and this idea is quite similar with a theorem in linear algebra called *Spectrum Theorem*.

As we proved before, generally observable must be a Hermitian, which, in this case, for simplicity, assume it is discrete, it have orthonormal eigen-wave-function $\varphi_1, \varphi_2, \cdots, \varphi_n$, with corresponding eigenvalue $f_1, f_2, \cdots, f_n$.

They satisfies the orthonormal condition

$$\int \varphi_n^* \varphi_m d\tau = \delta_{mn} \tag{3.5.1.1}$$

Within each state φ_n, particle have the definite value of f_n, we add these eigenstate together with some rate

$$\varphi = \sum_n C_n \varphi_n \tag{3.5.1.2}$$

where C_n are some number, with the superposition principle, φ is also a state. Within this state, some particle is in state φ_1 with value f_1, some in φ_2 with value f_2, and so on. The rate in each state is determined by the coefficient C_n. Thus

the average of such observable is

$$\begin{aligned}\overline{F} =& \frac{\int \varphi^* \hat{F} \varphi d\tau}{\int \varphi^* \varphi d\tau} = \frac{\int (\sum_n C_n \varphi_n)^* \hat{F} \sum_m C_m \varphi_m d\tau}{\int \varphi^* \varphi d\tau} \\ =& \frac{\sum_n \sum_m C_n^* C_m f_m \delta_{nm}}{\sum C_n^* C_n} = \frac{\sum_n C_n^* C_n f_n}{\sum C_n^* C_n}\end{aligned} \tag{3.5.1.3}$$

With the idea of finding average, $|C_n|^2 = C_n^* C_n$ is representing the number with value f_n. Thus, for orthonormal state

$$1 = \int \varphi^* \varphi \, d\tau = \sum_n C_n^* C_n \tag{3.5.1.4}$$

In such case, $|C_n|$ is the probability for a particle to have value f_n. As represent by a equation is

$$P(f_n) = |C_n|^2 = C_n^* C_n \tag{3.5.1.5}$$

Thus, when a particle is under the state described by wave function φ, the value corresponding to that observable can only be the eigenvalue corresponded. Also, every-time we measure the particle, we can only get one of the value within it's eigenvalue, each with probability $P = |C_n|^2$.

3.5.2 Continuous Spectrum and General Spectrum

We just discussed the spectrum in discrete condition, in real-world condition, there also exist continuous spectrum and spectrum with either continuous and discrete component.
(For derivation on these context, see Appendix H)

3.5.3 Third Postulate of Quantum Mechanics

In previous sections, we learned that the wave function provides the probability distribution for the eigenvalues of an observable in a quantum system—that is, the probability of finding a particular eigenvalue upon measurement. But what exactly happens when an eigenvalue is measured? In classical mechanics, a measurement does not affect the state of the system. In contrast, when a measurement yields an eigenvalue f_n of an observable $\hat{F}$, the system immediately collapses into the corresponding eigenstate φ_n, meaning that the system then possesses the definite value f_n. This change of state during measurement is known as *Reduction*.
From this observation, we can state the *Third Postulate of Quantum Mechanics*: Given that an observable F is represented by a linear Hermitian operator $\hat{F}$, its eigenstates can be chosen to form an orthonormal set:

$$\begin{aligned}&\hat{F}\varphi_n = f_n \varphi_n, \\ &\int \varphi_m^* \varphi_n \, d\tau = \delta_{mn}, \\ &\sum_n \varphi_n(\mathbf{x}) \varphi_n^*(\mathbf{x}') = \delta^{(3)}(\mathbf{x} - \mathbf{x}').\end{aligned} \tag{3.5.3.1}$$

For a normalized state ψ satisfying $\int \psi^* \psi \, d\tau = 1$, we can expand it in the eigenbasis of $\hat{F}$:

$$\begin{aligned}\psi &= \sum_n C_n \varphi_n, \\ C_n &= \int \varphi_n^* \psi \, d\tau.\end{aligned} \tag{3.5.3.2}$$

Here, C_n is the probability amplitude associated with the eigenvalue f_n, and the probability of measuring f_n is given by

$$P(f_n) = |C_n|^2 = \left| \int \varphi_n^* \psi \, d\tau \right|^2. \tag{3.5.3.3}$$

After the measurement yielding f_n, the state of the system collapses to the corresponding eigenstate φ_n.

3.6 State Space, Representation and Dirac Notation

3.6.1 State Space and Representation

Last section, we know that given a state ψ we can find out the distribution of any observables $\hat{F}$. In fact, our normal representation of an wave function $\psi(\mathbf{x}, t)$ is just a distribution of observable $\hat{\mathbf{x}}$. Expand this idea to other observables, we can say, any observable can be written as a wave function representing this state, which is a different form of the wave function, we call it as a different *Representation*.
In three dimensional space, any vector $\mathbf{R}$ can be de-composite into a superposition of three orthonormal basis $\hat{\mathbf{e}}$,

$$\begin{cases}\mathbf{R} = r_1 \hat{\mathbf{e}}_1 + r_2 \hat{\mathbf{e}}_2 + r_3 \hat{\mathbf{e}}_3 \\ \hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_j = \delta_{ij} \\ r_i = \hat{\mathbf{e}}_i \cdot \mathbf{R}, \quad i, j = 1, 2, 3\end{cases} \tag{3.6.1.1}$$

where r_i is the expansion coefficient, or coordinate.
Now the eigenstate φ of observable $\hat{F}$ form a complete coordinate where we derived last section (3.5.3.1).

$$\begin{cases}\hat{F}\varphi_n = f_n \varphi_n \\ \int \varphi_n^* \varphi_m \, dx \equiv (\varphi_n, \varphi_m) = \delta_{mn}\end{cases} \tag{3.6.1.2}$$

This form a representation of $\hat{F}$, with each state represented by a superposition of these eigenstates.

$$\begin{aligned}\varphi &= \sum_n C_n \varphi_n \\ C_n &= \int \varphi_n^* \varphi \, dx = (\varphi_n, \varphi)\end{aligned} \tag{3.6.1.3}$$

and C_n form a coordinate under this representation.
Thus, we can conclude that, as the eigenstate of observable $\hat{F}$ form a orthonormal basis for this state space, any state can be seen as a vector under this state space, called state vector. This state space we call it *Hilbert Space*. Any different form of wave function is actually a representation under different coordinate in Hilbert space.
As state is a vector in Hilbert space, we want to find a representation for operator too. Since linear operator is actually doing a linear transformation from a state to another state.

$$\varphi = \hat{G}\psi \tag{3.6.1.4}$$

where state ψ, and φ can both be represented under a complete orthonormal basis $\{\hat{F}, f_n, \varphi_n\}$.

$$\begin{cases}\psi = \sum_n c_n \varphi_n \\ \varphi = \sum_m b_m \varphi_m\end{cases} \tag{3.6.1.5}$$

Substitute back into (3.6.1.4)

$$\begin{aligned}\sum_m b_m \varphi_m &= \hat{G} \sum_n c_n \varphi_n = \sum_n c_n \hat{G} \varphi_n \\ b_m &= \sum c_n \int \varphi_m^* \hat{G} \varphi_n d\tau = \sum g_{mn} c_n\end{aligned} \tag{3.6.1.6}$$

which

$$g_{mn} = \int \varphi_m^* \hat{G} \varphi_n d\tau = (\varphi_m, \hat{G}\varphi_n) \tag{3.6.1.7}$$

Which is how we represent observable in F representation.

3.6.2 Representing by Matrices

There exist a more geometrical language of representing what we discussed in last section. For example, given a orthogonal basis $\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3$, and coordinate r_1, r_2, r_3. To represent a vector $\mathbf{R}$, we can arrange into a column vector

$$\mathbf{R} = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix} \tag{3.6.2.1}$$

And the linear transformation in between these vectors can be represented by a square matrix.
This is same in Hilbert Space, given a representation $F\{\hat{F}, f_n, \varphi_n\}$, using c_n to represent φ

$$\mathbf{C} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \end{pmatrix} \tag{3.6.2.2}$$

Thus the normalization condition $\int \varphi^* \phi d\tau = 1 = \sum |c_n|^2 = \sum c_n^* c_n$ can be represented by

$$1 = \begin{pmatrix} c_1^* & c_2^* & \cdots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \end{pmatrix} = \mathbf{C}^\dagger \mathbf{C} \tag{3.6.2.3}$$

Then $\hat{G}$ is

$$\mathbf{G} = (g_{mn}) = \begin{pmatrix} g_{11} & g_{12} & g_{13} & \cdots \\ g_{21} & g_{22} & g_{23} & \cdots \\ g_{31} & g_{32} & g_{33} & \cdots \\ \vdots & \vdots & \vdots & \end{pmatrix} \tag{3.6.2.4}$$

Thus the linear transformation between state φ, and ψ can be denoted as

$$\mathbf{B} = \mathbf{G}\mathbf{C} \tag{3.6.2.5}$$

And the Hermitian nature of operator tells us $g_{mn} = g_{mn}^*$, in which $\mathbf{G}$ is a Hermitian matrix

$$\mathbf{G} = \mathbf{G}^\dagger = \begin{pmatrix} g_{11}^* & g_{12}^* & g_{13}^* & \cdots \\ g_{21}^* & g_{22}^* & g_{23}^* & \cdots \\ g_{31}^* & g_{32}^* & g_{33}^* & \cdots \\ \vdots & \vdots & \vdots & \end{pmatrix} \tag{3.6.2.6}$$

The eigenvalue formula of operator $\hat{G}$ can also be represented as the eigenvalue formula of $\mathbf{G}$

$$\hat{G}\varphi = g\varphi \qquad (\mathbf{G} - g\mathbf{I})\mathbf{B} = 0 \tag{3.6.2.7}$$

For a special condition, operator $\hat{F}$ under it's own representation, every entry in the matrix

$$F_{mn} = \int \varphi_m^* \hat{F} \varphi_n d\tau = f_n \delta_{mn} \tag{3.6.2.8}$$

in which $\mathbf{F}$ is a diagonal matrix

$$F = \begin{pmatrix} f_1 & 0 & 0 & \cdots \\ 0 & f_2 & 0 & \cdots \\ \vdots & \vdots & \vdots & \\ 0 & 0 & 0 & f_n \end{pmatrix} \tag{3.6.2.9}$$

In linear algebra, we can find the eigenvalue of Hermitian matrix by diagonal using Unitary transformation. In application, actually, when we finding the energy level of hydrogen atom, is just finding the basis of energy representation.

3.6.3 Dirac Notation

As we brief introduced the vector's Dirac notation in section 2.1.1. A state vector can be denoted as

$$|\varphi\rangle \tag{3.6.3.1}$$

called *Ket* vector. Its Hermitian conjugate denoted

$$\langle\varphi| = (|\varphi\rangle)^\dagger \tag{3.6.3.2}$$

called the conjugate wave function, the *Bra* vector.
And define the inner product of $|\varphi\rangle$ and $|\psi\rangle$

$$\langle\varphi|\psi\rangle = \int \varphi^* \psi d\tau \tag{3.6.3.3}$$

Then we have a set of formulas we defined before, using Dirac notation can be denoted as

$$\begin{aligned} &|\varphi\rangle = \sum c_n |\varphi_n\rangle = \sum c_n |f_n\rangle \\ &\langle f_m | f_n\rangle = \delta_{mn} \\ &c_n = \langle f_n | \varphi\rangle \\ &|\varphi\rangle = \hat{G}|\psi\rangle \\ &\langle\varphi| = (\hat{G}|\varphi\rangle)^\dagger \equiv \langle\varphi|\hat{G}^\dagger \end{aligned} \tag{3.6.3.4}$$

And the normalization condition

$$\langle\varphi|\varphi\rangle = 1 \tag{3.6.3.5}$$

The definition of the Hermitian of an operator

$$\langle\varphi|\hat{G}\psi\rangle = \langle\hat{G}\varphi|\psi\rangle \equiv \langle\varphi|\hat{G}|\psi\rangle \tag{3.6.3.6}$$

The matrix entry of $\hat{G}$

$$g_{nm} = \langle f_n|\hat{G}|f_m\rangle \tag{3.6.3.7}$$

When representing with state vector, $|\varphi\rangle$ using a column matrix

$$\mathbf{C} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \end{pmatrix} = \begin{pmatrix} \langle f_1|\varphi\rangle \\ \langle f_2|\varphi\rangle \\ \vdots \end{pmatrix} \tag{3.6.3.8}$$

Its Hermitian conjugate

$$\mathbf{C} = \begin{pmatrix} c_1 & c_2 & \cdots \end{pmatrix} = \begin{pmatrix} \langle f_1|\varphi\rangle & \langle f_2|\varphi\rangle & \cdots \end{pmatrix} \tag{3.6.3.9}$$

And observable $\hat{G}$ using a square matrix

$$\mathbf{G} = (\langle f_m|\hat{G}|f_n\rangle) = \begin{pmatrix} \langle f_1|\hat{G}|f_1\rangle & \langle f_1|\hat{G}|f_2\rangle & \cdots \\ \langle f_2|\hat{G}|f_1\rangle & \langle f_2|\hat{G}|f_2\rangle & \cdots \\ \vdots & \vdots & \end{pmatrix} \tag{3.6.3.10}$$

The completeness condition for eigenstate

$$\begin{aligned} |\varphi\rangle &= \sum c_n |f_n\rangle = \sum |f_n\rangle \langle f_n|\varphi\rangle \\ &= \sum |f_n\rangle \langle f_n| \cdot |\varphi\rangle \end{aligned} \tag{3.6.3.11}$$

in which $\sum |f_n\rangle \langle f_n|$ is actually an operator, called identity operator, since it transform any state into it self

$$\sum |f_n\rangle \langle f_n| = 1 \tag{3.6.3.12}$$

Using Dirac Notation, we can easily transform to normal form of wave function

Since the eigenstate of position representation is δ function, then the state $|\varphi\rangle$ in position representation is

$$\langle \mathbf{x}, \varphi \rangle = \int \delta^{(3)}(\mathbf{x} - \mathbf{x}')\varphi(\mathbf{x}', t)d\tau' = \varphi(\mathbf{x}, t) \tag{3.6.3.13}$$

And thus, the momentum eigenstate $|\mathbf{p}\rangle$ in position representation is

$$\langle \mathbf{x}, \mathbf{p} \rangle = \int \delta^{(3)}(\mathbf{x}-\mathbf{x}')\frac{1}{(2\pi\hbar)^{3/2}}e^{i\mathbf{p}\cdot\mathbf{x}'/\hbar}d\tau' = \frac{1}{(2\pi\hbar)^{3/2}}e^{i\mathbf{p}\cdot\mathbf{x}'/\hbar} \tag{3.6.3.14}$$

Therefore, in momentum representation the state $|\varphi\rangle$ is represented as

$$\begin{aligned}\langle \mathbf{x}, \varphi \rangle &= \int \langle \mathbf{p}|\mathbf{x}\rangle \langle \mathbf{x}|\varphi\rangle \, d\tau' = \frac{1}{(2\pi\hbar)^{3/2}} \int e^{i\mathbf{p}\cdot\mathbf{x}'/\hbar}\varphi(\mathbf{x}, t)d\tau \\ &= C(\mathbf{p})\end{aligned} \tag{3.6.3.15}$$

3.6.4 Unitary Transformation

As we discussed in previous sections, when we know a wave function of a state, we can find the distribution of every observable. Since they are representing same state, what's the relationship between these different representation?
Under Hilbert Space, the answer is that this transformation actually a change of coordinate in the space.
Suppose we have two set of basis in Hilbert Space

$$\begin{aligned}&\{\hat{F}, f_n, |\psi_n\rangle\} \\ &\{\hat{G}, g_a, |\varphi_a\rangle\}\end{aligned} \tag{3.6.4.1}$$

By the completeness basis, the basis φ_a under G representation be represented by a basis ψ_m of F representation

$$\begin{cases} |\varphi_a\rangle = \sum\limits_n S_{na} |\psi_n\rangle \\ S_{na} = \langle \psi_n | \varphi_a \rangle \\ \langle \varphi_b | = \sum \langle \psi_m | \, S^*_{mb} \\ S^*_{mb} = \langle \psi_m | \varphi_b \rangle^* = \langle \varphi_b | \psi_m \rangle \end{cases} \tag{3.6.4.2}$$

Number S_{na} reflects the transformation between these two complete orthogonal basis. It should satisfy this condition

$$\begin{aligned}\delta_{ab} = \langle \varphi_a | \varphi_b \rangle &= \sum_{nm} S^*_{ma} S_{na} \langle \psi_m | \psi_n \rangle \\ &= \sum_{nm} S^*_{ma} S_{nb} \delta_{mn} = \sum_n S^*_{na} S_{nb}\end{aligned} \tag{3.6.4.3}$$

If we represent S_{na} as an matrix

$$\mathbf{S} = (S_{na}) = \begin{pmatrix} S_{11} & S_{12} & S_{13} & \cdots \\ S_{21} & S_{22} & S_{23} & \cdots \\ \vdots & \vdots & \vdots & \end{pmatrix} \tag{3.6.4.4}$$

And equation (3.6.4.2) can be denoted as

$$\begin{pmatrix} |\varphi_1\rangle \\ |\varphi_2\rangle \\ |\varphi_3\rangle \\ \vdots \end{pmatrix} = \begin{pmatrix} S_{11} & S_{21} & S_{31} & \cdots \\ S_{12} & S_{22} & S_{32} & \cdots \\ S_{13} & S_{23} & S_{33} & \cdots \\ \vdots & \vdots & \vdots & \end{pmatrix} \begin{pmatrix} |\psi_1\rangle \\ |\psi_2\rangle \\ |\psi_3\rangle \\ \vdots \end{pmatrix} = \mathbf{S}^T \begin{pmatrix} |\psi_1\rangle \\ |\psi_2\rangle \\ |\psi_3\rangle \\ \vdots \end{pmatrix} \tag{3.6.4.5}$$

And 3.6.4.3 can be represented by

$$\mathbf{S}^\dagger \mathbf{S} = \mathbf{I} \tag{3.6.4.6}$$

With completeness of basis,

$$\begin{aligned}(\mathbf{S}^\dagger \mathbf{S})_{nm} &= \sum_a S_{na} S^\dagger_{ma} \\ &= \sum_a \langle \psi_m | \varphi_a \rangle \langle \varphi_a | \psi_n \rangle \\ &= \langle \psi_n | \sum |\varphi_a\rangle \langle \varphi_a | \, |\psi_n\rangle \\ &= \langle \psi_n | \mathbf{I} | \psi_m \rangle = \delta_{mn} \\ \Rightarrow &\mathbf{S}\mathbf{S}^\dagger = \mathbf{I}\end{aligned} \tag{3.6.4.7}$$

From equation (3.6.4.6) and (3.6.4.7), it is clearly to see that $\mathbf{S}$ is an *Unitary Matrix*. Thus, the transformation between complete orthogonal basis is represented by unitary matrix, calling the unitary transformation of observables.

Note: In case of finite dimensional space, only (3.6.4.6) *can be the evidence of unitary; however, in case of infinite dimensional space,* (3.6.4.6) *and* (3.6.4.7) *should both satisfies to say* $\mathbf{S}$ *is unitary.*

Knowing that transformation is unitary, a further question is, how the wave function and observable transformed by this transformation.
Suppose a state in different representation

$$\begin{aligned}|\varphi\rangle &= \sum C^G_a \, |\varphi_a\rangle \\ |\varphi\rangle &= \sum C^F_n \, |\psi_n\rangle\end{aligned} \tag{3.6.4.8}$$

in which under F representation, the wave function is represented by C^F_n; under G representation, the wave function is represented by C^G_a. We can clearly set up a equality

$$|\varphi\rangle = \sum C^G_a \, |\varphi_a\rangle = \sum C^F_n \, |\psi_n\rangle \tag{3.6.4.9}$$

By (3.6.4.2)

$$\sum_a C^G_a \sum_m S_{ma} \, |\psi_m\rangle = \sum C^F_n \, |\psi_n\rangle$$

$$\sum C^G_a S_{ma} = C^F_m$$

or

$$\mathbf{C}^F = \mathbf{S}\mathbf{C}^G \tag{3.6.4.10}$$

and inversely

$$\mathbf{C}^G = \mathbf{S}^{-1}\mathbf{C}^F = \mathbf{S}^\dagger \mathbf{C}^F \tag{3.6.4.11}$$

Which is how wave function transformed under unitary transformation

Let's now continuing with how observable transformed under unitary transformation.
Suppose $\hat{H}$ changing $|A\rangle$ to $|B\rangle$, in which each represented in F representation as

$$\mathbf{A}^F = \begin{pmatrix} a^F_1 \\ a^F_2 \\ \vdots \end{pmatrix}, \quad \mathbf{B}^F = \begin{pmatrix} b^F_1 \\ b^F_2 \\ \vdots \end{pmatrix}, \quad \mathbf{H}^F = \begin{pmatrix} H^F_{11} & H^F_{12} & \cdots \\ H^F_{21} & H^F_{22} & \cdots \\ \vdots & \vdots & \end{pmatrix} \tag{3.6.4.12}$$

In G representation

$$\mathbf{A}^G = \begin{pmatrix} a^G_1 \\ a^G_2 \\ \vdots \end{pmatrix}, \quad \mathbf{B}^G = \begin{pmatrix} b^G_1 \\ b^G_2 \\ \vdots \end{pmatrix}, \quad \mathbf{H}^G = \begin{pmatrix} H^G_{11} & H^G_{12} & \cdots \\ H^G_{21} & H^G_{22} & \cdots \\ \vdots & \vdots & \end{pmatrix} \tag{3.6.4.13}$$

Thus, we have

$$\begin{aligned}\mathbf{B}^F &= \mathbf{H}^F\mathbf{A}^F \\ \mathbf{B}^G &= \mathbf{H}^G\mathbf{A}^G\end{aligned} \tag{3.6.4.14}$$

by the transformation of wave function,

$$\begin{aligned}\mathbf{B}^G &= \mathbf{S}^{-1}\mathbf{B}^F \\ \mathbf{A}^G &= \mathbf{S}^{-1}\mathbf{A}^F \\ \mathbf{B}^G &= \mathbf{S}^{-1}\mathbf{B}^F = \mathbf{S}^{-1}\mathbf{H}^F\mathbf{A}^F \\ &= \mathbf{H}^G\mathbf{A}^G = \mathbf{H}^G\mathbf{S}^{-1}\mathbf{A}^F \\ \mathbf{S}^{-1}\mathbf{H}^F\mathbf{A}^F &= \mathbf{H}^G\mathbf{S}^{-1}\mathbf{A}^F\end{aligned}$$

Since $\mathbf{A}$ is arbitrary,

$$\begin{aligned}\mathbf{S}^{-1}\mathbf{H}^F &= \mathbf{H}^G\mathbf{S}^{-1} \\ \mathbf{H}^G &= \mathbf{S}^{-1}\mathbf{H}^F\mathbf{S}\end{aligned} \tag{3.6.4.15}$$

this equation also shown us the observable in different representation is similar, which can be obtained by a change of coordinate matrix, in this case, is the unitary matrix.

The unitary nature of transformations reflects the relationship between different representations of quantum states and observables. In physics, this means that a state remains essentially unchanged when we change the "language" or basis used to describe it. Below are some examples of properties preserved by unitary transformations:

(1) *Preservation of the Inner Product:*

The inner product between two quantum states quantifies their similarity. Unitary transformations preserve inner products, so the relationship between states—particularly, the orthogonality of distinct states—remains unchanged under a unitary change of basis.

(2) *Preservation of Eigenvalues and Trace:*

Algebraically, a unitary transformation leaves the eigenvalues of an operator unchanged. Physically, the eigenvalues of an observable, which represent measurable quantities, should remain invariant under a change of representation. Similarly, the trace of an operator, which is the sum of its eigenvalues, is also invariant under unitary transformations.

(3) *Preservation of Matrix Elements:*

For any operator $\hat{F}$, the matrix element $\langle\varphi|\hat{F}|\psi\rangle$, which can be interpreted as the probability amplitude for the transition between the states $|\psi\rangle$ and $|\varphi\rangle$ under the action of $\hat{F}$, remains unchanged under a unitary transformation.

3.7 Basic Operators

3.7.1 Energy Operator

Previously, we have introduced position operator $\hat{x}$ and momentum operator $\hat{p}$. Another important operator in quantum mechanics is the energy operator, known as *Hamiltonian*. We will introduce it in this section.
By momentum operator and the definition of kinetic energy, we can get the kinetic energy operator

$$\hat{T} = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2}{2m}\nabla^2 \tag{3.7.1.1}$$

its eigen-wave-function is the wave function of free particle.

In spherical coordinate, it can be written as

$$\begin{aligned}\hat{T} =& \frac{-\hbar^2}{2mr^2}\left[\frac{\partial}{\partial r}r^2\frac{\partial}{\partial r} + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right] \\ =& -\frac{\hbar^2}{2mr^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\hat{L}^2}{2mr^2}\end{aligned} \tag{3.7.1.2}$$

Generally speaking, any particle under a conservative energy field, the energy is the addition of kinetic energy and potential energy which is given by

$$\hat{H} = \hat{T} + \hat{U} \tag{3.7.1.3}$$

In classical mechanics, this is called a Hamiltonian function, and thus, in quantum mechanics, we call this operator as *Hamiltonian Operator*.
Different with position and momentum operator, the eigenvalue and eigen-wave-function is different with the change of the system, meaning that it reflects the specific property of a system, and thus, it is important for our study.

4 Classic Models and Applications

4.1 One-Dimensional Infinite Square Well

4.1.1 General Approach for this Model

Consider a particle confined in an infinite square well of width L. The potential energy is defined as (as shown in Figure 3)

$$V(x) = \begin{cases} 0, & 0 < x < L, \\ \infty, & \text{otherwise.} \end{cases} \tag{4.1.1.1}$$

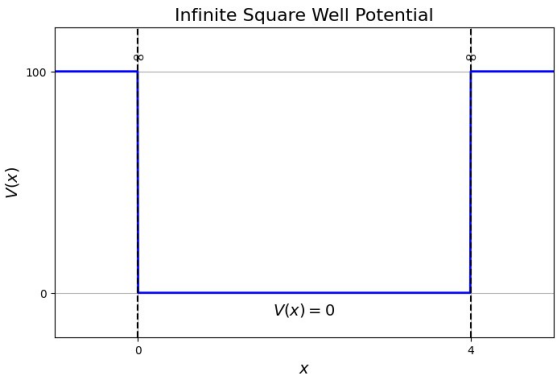


Figure 3: Infinite Square Well

Since the potential is infinite outside the interval $(0, L)$, the particle is strictly confined within the well. Therefore, the wave function must vanish outside the well:

$$\psi(x) = 0 \quad \text{for } x \le 0 \text{ and } x \ge L. \tag{4.1.1.2}$$

Within the well, where $V(x) = 0$, the time-independent Schrödinger equation becomes

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} = E\psi(x). \tag{4.1.1.3}$$

It is convenient to rewrite this equation as

$$\frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0, \tag{4.1.1.4}$$

where

$$k = \sqrt{\frac{2mE}{\hbar^2}}. \tag{4.1.1.5}$$

The general solution to this second-order differential equation is

$$\psi(x) = A\sin(kx) + B\cos(kx), \tag{4.1.1.6}$$

where A and B are constants to be determined by the boundary conditions.

Application of Boundary Conditions

1. At $x = 0$:

   The boundary condition $\psi(0) = 0$ gives

   $$\psi(0) = A\sin(0) + B\cos(0) = B = 0. \tag{4.1.1.7}$$

   Hence, $B = 0$ and the solution simplifies to

   $$\psi(x) = A\sin(kx). \tag{4.1.1.8}$$

2. At $x = L$:

   The boundary condition $\psi(L) = 0$ requires

   $$\psi(L) = A\sin(kL) = 0. \tag{4.1.1.9}$$

   For a non-trivial solution (i.e., $A \neq 0$), we must have

   $$\sin(kL) = 0. \tag{4.1.1.10}$$

   This condition is satisfied when

   $$kL = n\pi, \quad \text{where } n = 1, 2, 3, \ldots \tag{4.1.1.11}$$

   Thus, the allowed values of k are

   $$k_n = \frac{n\pi}{L}. \tag{4.1.1.12}$$

Energy Eigenvalues and Normalization

Substituting k_n into the relation $k = \sqrt{\frac{2mE}{\hbar^2}}$, the energy eigenvalues are found to be

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2\pi^2\hbar^2}{2mL^2}, \quad n = 1, 2, 3, \ldots \tag{4.1.1.13}$$

The normalized eigenfunctions must satisfy

$$\int_0^L |\psi_n(x)|^2\, dx = 1. \tag{4.1.1.14}$$

Given

$$\psi_n(x) = A\sin\left(\frac{n\pi x}{L}\right), \tag{4.1.1.15}$$

we compute

$$\int_0^L A^2 \sin^2\left(\frac{n\pi x}{L}\right) dx = A^2\frac{L}{2} = 1. \tag{4.1.1.16}$$

Solving for A yields

$$A = \sqrt{\frac{2}{L}}. \tag{4.1.1.17}$$

Thus, the normalized eigenfunctions are

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right), \quad n = 1, 2, 3, \ldots \tag{4.1.1.18}$$

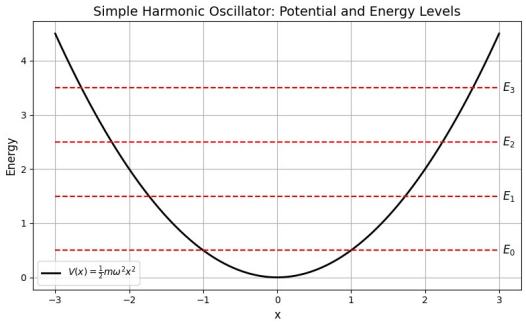


Figure 4: The Potential Of Simple Harmonic Oscillator and its Energy Level

4.2 Simple Harmonic Oscillator

4.2.1 General Approach for this model

Suppose a particle is in a potential (Figure 4) given by

$$V(x) = \frac{1}{2}m\omega^2x^2, \tag{4.2.1.1}$$

and the time-independent Schrödinger equation for the oscillator is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \frac{1}{2}m\omega^2x^2\psi(x) = E\psi(x). \tag{4.2.1.2}$$

To simplify the equation, we introduce a dimensionless variable:

$$\xi = \sqrt{\frac{m\omega}{\hbar}}\, x. \tag{4.2.1.3}$$

Then,

$$\frac{d}{dx} = \sqrt{\frac{m\omega}{\hbar}}\,\frac{d}{d\xi}, \quad \frac{d^2}{dx^2} = \frac{m\omega}{\hbar}\,\frac{d^2}{d\xi^2}.$$

Substituting these into the Schrödinger equation

$$-\frac{\hbar^2}{2m}\left(\frac{m\omega}{\hbar}\frac{d^2\psi}{d\xi^2}\right) + \frac{1}{2}m\omega^2\left(\frac{\hbar}{m\omega}\xi^2\right)\psi = E\psi, \tag{4.2.1.4}$$

which simplifies to

$$-\frac{\hbar\omega}{2}\frac{d^2\psi}{d\xi^2} + \frac{\hbar\omega}{2}\xi^2\psi = E\psi. \tag{4.2.1.5}$$

Dividing both sides by $\hbar\omega$ gives

$$-\frac{1}{2}\frac{d^2\psi}{d\xi^2} + \frac{1}{2}\xi^2\psi = \frac{E}{\hbar\omega}\psi. \tag{4.2.1.6}$$

Defining the dimensionless energy $\epsilon = \frac{E}{\hbar\omega}$, we have

$$\frac{d^2\psi}{d\xi^2} = \left(\xi^2 - 2\epsilon\right)\psi. \tag{4.2.1.7}$$

For large $|\xi|$, the term ξ^2 dominates, so the differential equation approximates to

$$\frac{d^2\psi}{d\xi^2} \approx \xi^2\psi.$$

The asymptotic solution of this equation behaves as $\psi(\xi) \sim e^{-\xi^2/2}$ (we discard the exponentially growing solution). Hence, we write

$$\psi(\xi) = e^{-\xi^2/2}h(\xi), \tag{4.2.1.8}$$

where $h(\xi)$ is a function to be determined.
Substituting $\psi(\xi) = e^{-\xi^2/2}h(\xi)$ into the full differential equation, we compute

$$\frac{d\psi}{d\xi} = e^{-\xi^2/2}\frac{dh}{d\xi} - \xi e^{-\xi^2/2}h(\xi),$$

$$\frac{d^2\psi}{d\xi^2} = e^{-\xi^2/2}\frac{d^2h}{d\xi^2} - 2\xi e^{-\xi^2/2}\frac{dh}{d\xi} + (\xi^2-1)e^{-\xi^2/2}h(\xi).$$

Plugging into the Schrödinger equation:

$$-\frac{1}{2}\left[e^{-\xi^2/2}\frac{d^2h}{d\xi^2} - 2\xi e^{-\xi^2/2}\frac{dh}{d\xi} + (\xi^2-1)e^{-\xi^2/2}h(\xi)\right] + \frac{1}{2}\xi^2 e^{-\xi^2/2}h(\xi) = \epsilon\, e^{-\xi^2/2}h(\xi).$$

After canceling the common factor $e^{-\xi^2/2}$ and simplifying, we obtain:

$$-\frac{1}{2}\frac{d^2h}{d\xi^2} + \xi\frac{dh}{d\xi} + \frac{1}{2}h(\xi) = \epsilon\, h(\xi).$$

Multiplying both sides by 2 gives:

$$-\frac{d^2h}{d\xi^2} + 2\xi\frac{dh}{d\xi} + h(\xi) = 2\epsilon\, h(\xi). \tag{4.2.1.9}$$

Rearrange to:

$$\frac{d^2h}{d\xi^2} - 2\xi\frac{dh}{d\xi} + (2\epsilon - 1)\, h(\xi) = 0. \tag{4.2.1.10}$$

This is recognized as Hermite's differential equation. The polynomial solutions $h(\xi) = H_n(\xi)$ occur when

$$2\epsilon - 1 = 2n, \quad n = 0, 1, 2, \ldots, \tag{4.2.1.11}$$

so that

$$\epsilon = n + \frac{1}{2}. \tag{4.2.1.12}$$

Thus, the energy eigenvalues are

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right). \tag{4.2.1.13}$$

The corresponding eigenfunctions are given by

$$\psi_n(x) = N_n\, e^{-m\omega x^2/(2\hbar)}\, H_n\left(\sqrt{\frac{m\omega}{\hbar}}\, x\right), \tag{4.2.1.14}$$

where $H_n(\xi)$ are the Hermite polynomials and N_n is a normalization constant determined by

$$N_n = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\frac{1}{\sqrt{2^n n!}}. \tag{4.2.1.15}$$

4.2.2 Approaching with Representation Theory

In section 3 we discussed operators and different representations, how do we apply them in application? Now let's solve this problem with representation theory.

Emission and Absorption Operator

The Hamiltonian of 1-dimensional simple harmonic oscillator is

$$\hat{H} = \frac{1}{2\mu}\hat{p}^2 + \frac{1}{2}k\hat{x}^2 = \frac{1}{2\mu}\hat{p}^2 + \frac{1}{2}\mu\omega^2\hat{x}^2$$
$$\omega = \sqrt{\frac{k}{\mu}} \tag{4.2.2.1}$$

Now we introduce a new operator

$$\hat{a} = \left(\frac{\mu\omega}{2\hbar}\right)^{1/2}\left(\hat{x} + \frac{i}{\mu\omega}\hat{p}\right) \tag{4.2.2.2}$$

its Hermitian conjugate

$$\hat{a}^\dagger = \left(\frac{\mu\omega}{2\hbar}\right)^{1/2}\left(\hat{x} - \frac{i}{\mu\omega}\hat{p}\right) \tag{4.2.2.3}$$

We can also write $\hat{x}$ and $\hat{p}$ in the form

$$\begin{cases}\hat{x} = \left(\dfrac{\hbar}{2\mu\omega}\right)^{1/2}(\hat{a} + \hat{a}^\dagger) \\ \hat{p} = \dfrac{1}{i}\left(\dfrac{\hbar\mu\omega}{2}\right)^{1/2}(\hat{a} - \hat{a}^\dagger)\end{cases} \tag{4.2.2.4}$$

It is clearly to see that $\hat{a}$ is not a Hermitian operator as it does not have physical meaning. Using the commute relation between $\hat{x}$ and $\hat{p}$ can see the commute relation bewteen $\hat{a}$ and $\hat{a}^\dagger$ is

$$\begin{aligned}[\hat{a}, \hat{a}^\dagger] =& \frac{\mu\omega}{2\hbar}\left[\hat{x} + \frac{i}{\mu\omega}\hat{p}, \hat{x} - \frac{i}{\mu\omega}\hat{p}\right] \\ =& \frac{\mu\omega}{2\hbar}\left[\hat{x}, \frac{-i}{\mu\omega}\hat{p}\right] + \left[\frac{-i}{\mu\omega}\hat{p}, \hat{x}\right] \\ =& 1\end{aligned} \tag{4.2.2.5}$$

In such case, Hamiltonian can be rewrite as

$$\begin{aligned}\hat{H} =& \frac{1}{2\mu}(\hat{p} + \mu^2\omega^2\hat{x}^2) = \frac{\hbar\omega}{4}[(\hat{a} + \hat{a}^\dagger)^2 - (\hat{a} - \hat{a}^\dagger)^2] \\ =& \frac{\hbar\omega}{4}(2\hat{a}^\dagger\hat{a} + 1) = \hbar\omega\hat{a}^\dagger\hat{a} + \frac{1}{2}\hbar\omega \\ =& \hbar\omega\left(\hat{N} + \frac{1}{2}\right), \qquad \hat{N} = \hat{a}^\dagger\hat{a}\end{aligned} \tag{4.2.2.6}$$

where $\hat{N}$ is called particle number operator, and its Hermitian.

Eigenstate of Particle Number Operator

As we can see in (4.2.2.6), to find the eigenvalue of this Hamiltonian, we can first find the eigenvalue of the particle number operator.
First, intuitively, we can see, the eigenvalue of $\hat{N}$ should be non-negative, let's set the eigenstate of particle number as $|n\rangle$

$$\begin{cases}\hat{N}\,|n\rangle = n\,|n\rangle \\ \langle n|n\rangle = 1\end{cases} \tag{4.2.2.7}$$

algebraically $n \geq 0$ because

$$\langle n|\hat{N}|n\rangle = n\,\langle n|n\rangle = n$$

and

$$\langle n|\hat{N}|n\rangle = \langle n|\hat{a}^\dagger\hat{a}|n\rangle$$

By definition, this is the square of mod of $\hat{a}\,|n\rangle$, so must be non-negative, which

$$\langle n|\hat{N}|n\rangle \geq 0 \tag{4.2.2.8}$$

and the equality in (4.2.2.8) satisfies if and only if $\hat{a}\,|n\rangle = 0$
We can thus conclude that the energy level of simple harmonic oscillator obey this relation

$$E_n = \hbar\omega n + \frac{1}{2}\hbar\omega \geq \frac{1}{2}\hbar\omega \tag{4.2.2.9}$$

The commute relationship between $\hat{a}$ and $\hat{N}$ is

$$[\hat{a}, \hat{N}] = [\hat{a}, \hat{a}^\dagger \hat{a}] = \hat{a}\,, \quad [\hat{a}^\dagger, \hat{N}] = -\hat{a}^\dagger \tag{4.2.2.10}$$

Then we have

$$\begin{aligned} \hat{N}\hat{a}\,|n\rangle =& ([\hat{N}, \hat{a}] + \hat{a}\hat{N})\,|n\rangle \\ =& (n-1)\hat{a}\,|n\rangle \end{aligned} \tag{4.2.2.11}$$

$$\begin{aligned} \hat{N}\hat{a}^\dagger\,|n\rangle =& ([\hat{N}, \hat{a}^\dagger] + \hat{a}^\dagger\hat{N})\,|n\rangle \\ =& (n+1)\hat{a}^\dagger\,|n\rangle \end{aligned} \tag{4.2.2.12}$$

This means that, if $|n\rangle$ is eigenvector of eigenvalue of $\hat{N}$, then $\hat{a}\,|n\rangle$ is also eigenvector of $\hat{N}$ with eigenvalue $(n-1)$, and so for $\hat{a}^\dagger\,|n\rangle$ with eigenvalue $(n+1)$
We can conclude with a sequence of eigenvalue and eigenvector

$$\begin{aligned} \text{Eigenvector} \quad & \cdots, \hat{a}^2\,|n\rangle, \hat{a}\,|n\rangle, |n\rangle, \hat{a}^\dagger\,|n\rangle, \hat{a}^{\dagger 2}\,|n\rangle, \cdots \\ \text{Eigenvalue} \quad & \cdots, n-2,\ n-1,\ n, n+1,\ n+2, \cdots \end{aligned} \tag{4.2.2.13}$$

But since the eigenvalue can not be negative, we set the smallest eigenvalue n_0, with corresponded eigenvector $|n_0\rangle$

$$\begin{aligned} &\hat{N}\,|n_0\rangle = n_0\,|n_0\rangle \\ &\langle n_0|n_0\rangle = 1 \\ &\hat{a}\,|n_0\rangle = 0 \end{aligned} \tag{4.2.2.14}$$

Then we have

$$n_0 = \langle n_0|\hat{N}|n_0\rangle = \langle n_0|\hat{a}^\dagger\hat{a}|n_0\rangle = 0 \tag{4.2.2.15}$$

and obtain a new sequence

$$\begin{aligned} \text{Eigenvector} \quad & |0\rangle, |1\rangle, |2\rangle, \cdots, |n\rangle, \cdots \\ \text{Eigenvalue} \quad & 0,\ 1,\ 2,\ \cdots,\ n,\ \cdots \end{aligned} \tag{4.2.2.16}$$

To normalize these eigenvector, first set the lowest state $|0\rangle$ is normalized

$$\langle 0|0\rangle = 1 \tag{4.2.2.17}$$

Then

$$\begin{aligned} \langle 0|\hat{a}\hat{a}^\dagger|0\rangle =& \langle 0|([\hat{a}, \hat{a}^\dagger] + \hat{a}^\dagger\hat{a})\rangle = 1 \\ \langle 0|\hat{a}\hat{a}\hat{a}^\dagger\hat{a}^\dagger|0\rangle =& \langle 0|\hat{a}(1+\hat{N})\hat{a}^\dagger|0\rangle \\ =& \langle 0|\hat{a}\hat{a}^\dagger|0\rangle + \langle 0|\hat{a}\hat{N}\hat{a}^\dagger|0\rangle \\ =& \langle 0|\hat{a}\hat{a}^\dagger|0\rangle + \langle 0|(\hat{N}\hat{a} + \hat{a})\hat{a}^\dagger|0\rangle \\ =& 2\,\langle 0|\hat{a}\hat{a}^\dagger|0\rangle + \langle 0|\hat{N}\hat{a}\hat{a}^\dagger|0\rangle = 2! \end{aligned}$$

Inductively we can see

$$\langle 0|\hat{a}^k\hat{a}^{\dagger k}|0\rangle = k! \tag{4.2.2.18}$$

Thus we can take normalized eigenvector as $|k\rangle = \dfrac{1}{\sqrt{k!}}\hat{a}^{\dagger k}\,|0\rangle$
Then

$$|n\rangle = \frac{1}{\sqrt{n!}}\hat{a}^{\dagger n}\,|0\rangle \tag{4.2.2.19}$$

$$\begin{cases} \hat{N}\,|n\rangle = n\,|n\rangle \\ \langle n|k\rangle = \delta_{nk} \\ n, k = 0, 1, 2, 3, \cdots \end{cases} \tag{4.2.2.20}$$

Matrix form of Operators

As we can form a representation with eigenvectors $|n\rangle$, the $\hat{N}$, with its eigenvalues, can form a matrix under this representation

$$N = \begin{pmatrix} 0 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 2 & 0 & \cdots \\ 0 & 0 & 0 & 3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \end{pmatrix} \tag{4.2.2.21}$$

And with entry $\langle m|a^\dagger|n\rangle$

$$\mathbf{a}^\dagger = \begin{pmatrix} 0 & 0 & 0 & \cdots \\ \sqrt{1} & 0 & 0 & \cdots \\ 0 & \sqrt{2} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} 0 & \sqrt{1} & 0 & \cdots \\ 0 & 0 & \sqrt{2} & \cdots \\ 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix} \tag{4.2.2.22}$$

Let's step back to see the energy level, since the energy operator and particle number operator is in linear relationship, the eigenvalue will be

$$E_n = \langle n|\hat{H}|n\rangle = \langle n|\hbar\omega\left(\hat{N} + \frac{1}{2}\right)|n\rangle = \left(n + \frac{1}{2}\right)\hbar\omega \tag{4.2.2.23}$$

Thus

$$\mathbf{H} = \begin{pmatrix} \hbar\omega/2 & 0 & 0 & \cdots \\ 0 & 3\hbar\omega/2 & 0 & \cdots \\ 0 & 0 & 5\hbar\omega/2 & \cdots \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix} \tag{4.2.2.24}$$

Where we have position operator in the form

$$\hat{x} = \left(\frac{\hbar}{2\mu\omega}\right)^{1/2}(\hat{a} + \hat{a}^\dagger) = \left(\frac{\hbar}{2\mu\omega}\right)^{1/2}\begin{pmatrix} 0 & \sqrt{1} & 0 & \cdots \\ \sqrt{1} & 0 & \sqrt{2} & \cdots \\ 0 & \sqrt{2} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix} \tag{4.2.2.25}$$

$$\begin{aligned} \hat{p} =& \frac{1}{i}\left(\frac{\hbar\mu\omega}{2}\right)^{1/2}(\hat{a} - \hat{a}^\dagger) \\ =& \frac{1}{i}\left(\frac{\hbar\mu\omega}{2}\right)^{1/2}\begin{pmatrix} 0 & \sqrt{1} & 0 & \cdots \\ -\sqrt{1} & 0 & \sqrt{2} & \cdots \\ 0 & -\sqrt{2} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix} \end{aligned} \tag{4.2.2.26}$$

Wave function of SHO

Now we already have the eigenstate of energy, and we want to find the wave function of each state, in another word, the position representation of energy eigenstate.
The state $|n\rangle$ with energy eigenvalue $(n+1/2)\hbar\omega$ satisfies the formula

$$\hat{H}\,|n\rangle = \frac{2n+1}{2}\hbar\omega\,|n\rangle \tag{4.2.2.27}$$

In position representation, the form

$$\hat{H}(\hat{x}, \hat{p})\,\langle x|n\rangle = \frac{2n+1}{2}\hbar\omega\,\langle x|n\rangle \tag{4.2.2.28}$$

which $\langle x|n\rangle$ is the position representation of wave function, denoted $\psi_n(x)$

$$\hat{H}(\hat{x}, \hat{p})\psi_n(x) = \frac{2n+1}{2}\hbar\omega\psi_n(x)$$

or

$$\left(\frac{\hat{p}}{2\mu}+\frac{1}{2}\mu\omega^2\hat{x}^2\right)\psi_n(x)=\left(n+\frac{1}{2}\right)\hbar\omega\psi_n(x)$$

substitute $\hat{p}$ into

$$\frac{\hbar^2}{2\mu}\frac{d^2\psi_n}{dx^2}+\left[\left(n+\frac{1}{2}\right)\hbar\omega-\frac{\mu\omega^2}{2}x^2\right]\psi_n=0 \tag{4.2.2.29}$$

We have already discussed the solution to this in section 4.2.1. And we can also get the solution by another approach:
As we obtained a boundary condition in ground state

$$\hat{a}\left|0\right\rangle=0 \tag{4.2.2.30}$$

as represented in position representation

$$\left\langle x\right|\hat{a}\left|0\right\rangle=0 \tag{4.2.2.31}$$

substitute the basis of position representation

$$\int\left\langle x|\hat{a}|x'\right\rangle\left\langle x'\right|\left|0\right\rangle dx'=0 \tag{4.2.2.32}$$

with the definition of absorption operator and momentum operator

$$\left(\frac{\mu\omega}{2\hbar}\right)^{1/2}\int\left\langle x\middle|\hat{x}+\frac{i}{\mu\omega}\hat{p}\middle|x'\right\rangle\left\langle x'|0\right\rangle dx' \tag{4.2.2.33}$$

or

$$\left(x+\frac{i}{\mu\omega}\frac{\hbar}{i}\frac{d}{dx}\right)\psi_0(x)=0 \tag{4.2.2.34}$$

in which the ground state wave function

$$\psi_0(x)=Ae^{\frac{1}{2}\frac{\mu\omega}{\hbar}x^2} \tag{4.2.2.35}$$

Similarly get the wave function of first excited state

$$\begin{aligned}\langle x|1\rangle=&\langle x|\hat{a}^\dagger|0\rangle\\=&\int\langle x|\hat{a}^\dagger|x'\rangle\langle x'|0\rangle\,dx\\=&\int\sqrt{\frac{\mu\omega}{2\hbar}}\left\langle x\right|\left(\hat{x}-\frac{i}{\mu\omega}\hat{p}\right)\left|x'\right\rangle\psi_0(x)\,dx'\\=&\sqrt{\frac{\mu\omega}{2\hbar}}\left(x-\frac{\hbar}{\mu\omega}\frac{d}{dx}\right)\psi_0(x)\\=&\sqrt{2}\alpha x\psi_0(x)\end{aligned} \tag{4.2.2.36}$$

Apply Heisenberg Uncertainty Relation to find Ground State Energy of SHO

An application of uncertainty relation in SHO is to find the ground state energy.
The Hamiltonian of SHO

$$\hat{H}=\frac{\hat{p}}{2\mu}+\frac{1}{2}\mu\omega^2\hat{x}^2 \tag{4.2.2.37}$$

the average energy in any state can be given by

$$\begin{aligned}E=&\overline{H}=\frac{1}{2\mu}\overline{p^2}+\frac{1}{2}\mu\omega^2\overline{x^2}\geq 2\left(\frac{1}{2\mu}\overline{p^2}\cdot\frac{1}{2}\mu\omega^2\overline{x^2}\right)^{1/2}\\=&\omega\left[\overline{p^2}\cdot\overline{x^2}\right]^{1/2}\geq\omega\left(\frac{1}{4}i\overline{[\hat{p},\hat{x}]^2}\right)^{1/2}=\frac{\hbar\omega}{2}\end{aligned} \tag{4.2.2.38}$$

which is the ground state energy of SHO.

Note: The first line of equation (4.2.2.38) is given by the in equality between algebraic and geometrical average $A+B\geq 2\sqrt{AB}$, where $A=\frac{1}{2\mu}\overline{p^2}$, $B=\frac{1}{2}\mu\omega^2\overline{x^2}$.

4.3 Three Dimensional Infinite Well

The three-dimensional infinite potential well (or box) is a natural extension of the one-dimensional infinite square well. Consider a particle confined within a rectangular box with dimensions L_x, L_y, and L_z. The potential is defined as

$$V(x,y,z)=\begin{cases}0, & 0<x<L_x,\ 0<y<L_y,\ 0<z<L_z,\\ \infty, & \text{otherwise.}\end{cases} \tag{4.3.1}$$

Inside the box, the time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}\right)=E\psi. \tag{4.3.2}$$

In such case, our idea is to use *Separation of Variables*

$$\psi=X(x)Y(y)Z(z) \tag{4.3.3}$$

substitute back into the Schrödinger equation

$$\begin{aligned}&X'(x)Y(y)Z(z)+X(x)Y'(y)Z(z)+X(x)Y(y)Z'(z)\\&+\frac{2mE}{\hbar^2}X(x)Y(y)Z(z)=0\end{aligned} \tag{4.3.4}$$

$$\begin{cases}\dfrac{X'(x)}{X(x)}+\dfrac{2mE_x}{\hbar^2}=0\\ \dfrac{Y'(x)}{Y(x)}+\dfrac{2mE_y}{\hbar^2}=0\\ \dfrac{Z'(x)}{Z(x)}+\dfrac{2mE_z}{\hbar^2}=0\end{cases}\quad\Rightarrow\quad\begin{cases}k_x=\dfrac{\sqrt{2mE_x}}{\hbar}\\ k_y=\dfrac{\sqrt{2mE_y}}{\hbar}\\ k_z=\dfrac{\sqrt{2mE_z}}{\hbar}\end{cases} \tag{4.3.5}$$

$$\begin{cases}\psi(x)=A_x\sin(k_xx+\theta_x)\\ \psi(y)=A_x\sin(k_yy+\theta_y)\\ \psi(z)=A_x\sin(k_zz+\theta_z)\end{cases} \tag{4.3.6}$$

Apply the boundary condition

$$\begin{cases}\psi_x(0)=A_x\sin(\theta_x)=0\\ \psi_y(0)=A_y\sin(\theta_y)=0\\ \psi_z(0)=A_z\sin(\theta_z)=0\end{cases}\quad\Rightarrow\quad\begin{cases}\theta_x=0\\ \theta_y=0\\ \theta_z=0\end{cases} \tag{4.3.7}$$

$$\begin{cases}\psi_x(L_x)=A_x\sin(\theta_x)=0\\ \psi_y(L_y)=A_x\sin(\theta_x)=0\\ \psi_z(L_z)=A_x\sin(\theta_x)=0\end{cases}\quad\Rightarrow\quad\begin{cases}k_x=\dfrac{n_x\pi}{L_x}\\ k_y=\dfrac{n_y\pi}{L_y}\\ k_z=\dfrac{n_z\pi}{L_z}\end{cases} \tag{4.3.8}$$

After normalization is

$$\begin{cases}\psi(x)=\sqrt{\dfrac{2}{L_x}}\sin(\dfrac{n_x\pi}{L_x}x)\\ \psi(y)=\sqrt{\dfrac{2}{L_y}}\sin(\dfrac{n_y\pi}{L_y}y)\\ \psi(z)=\sqrt{\dfrac{2}{L_z}}\sin(\dfrac{n_z\pi}{L_z}z)\end{cases} \tag{4.3.9}$$

And finally

$$\psi(x,y,z)=\sqrt{\frac{8}{L_xL_yL_z}}\sin(\frac{n_x\pi}{L_x}x)\sin(\frac{n_y\pi}{L_y}y)\sin(\frac{n_z\pi}{L_z}z) \tag{4.3.10}$$

With energy

$$\begin{aligned}E=&E_x+E_y+E_z\\=&\frac{k_x^2\hbar^2}{2m}+\frac{k_y^2\hbar^2}{2m}+\frac{k_z^2\hbar^2}{2m}\\=&\frac{\hbar}{2m}\left(\frac{n_x^2\pi}{L_x^2}+\frac{n_y^2\pi}{L_y^2}+\frac{n_z^2\pi}{L_z^2}\right)\end{aligned} \tag{4.3.11}$$

Let's analyze the situation when $L_x = L_y = L_z$, meaning the potential is a cube.
Below is the summarize the eigenvalues and degeneracies for the lowest few energy levels:

- Ground State:

$$n_x = n_y = n_z = 1, \quad E_0 = \frac{3\hbar^2\pi^2}{2mL^2},$$

$$\psi_{1,1,1}(x,y,z) = \sqrt{\frac{8}{L^3}} \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi y}{L}\right) \sin\left(\frac{\pi z}{L}\right).$$

  The ground state is non-degenerate.

- First Excited State: The first excited energy corresponds to the quantum numbers where one of them equals 2 and the other two equal 1. There are three possible combinations:

$$\begin{cases} n_x = 1,\ n_y = 1,\ n_z = 2, \\ n_x = 1,\ n_y = 2,\ n_z = 1, \\ n_x = 2,\ n_y = 1,\ n_z = 1. \end{cases}$$

  For each, the energy is

$$E_1 = \frac{\hbar^2\pi^2}{2mL^2}(1^2 + 1^2 + 2^2) = \frac{6\hbar^2\pi^2}{2mL^2}.$$

  Thus, the first excited state is threefold degenerate, with eigenfunctions:

$$\psi_{1,1,2}(x,y,z) = \sqrt{\frac{8}{L^3}} \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi y}{L}\right) \sin\left(\frac{2\pi z}{L}\right),$$

$$\psi_{1,2,1}(x,y,z) = \sqrt{\frac{8}{L^3}} \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{2\pi y}{L}\right) \sin\left(\frac{\pi z}{L}\right),$$

$$\psi_{2,1,1}(x,y,z) = \sqrt{\frac{8}{L^3}} \sin\left(\frac{2\pi x}{L}\right) \sin\left(\frac{\pi y}{L}\right) \sin\left(\frac{\pi z}{L}\right).$$

- Second Excited State: Here, we consider combinations with two quantum numbers equal to 2 and one equal to 1:

$$\begin{cases} n_x = 1,\ n_y = 2,\ n_z = 2, \\ n_x = 2,\ n_y = 1,\ n_z = 2, \\ n_x = 2,\ n_y = 2,\ n_z = 1. \end{cases}$$

  The energy for these states is

$$E_2 = \frac{\hbar^2\pi^2}{2mL^2}(1^2 + 2^2 + 2^2) = \frac{9\hbar^2\pi^2}{2mL^2},$$

  resulting in a threefold degeneracy, with eigenfunctions:

$$\psi_{1,2,2}(x,y,z) = \sqrt{\frac{8}{L^3}} \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{2\pi y}{L}\right) \sin\left(\frac{2\pi z}{L}\right),$$

$$\psi_{2,1,2}(x,y,z) = \sqrt{\frac{8}{L^3}} \sin\left(\frac{2\pi x}{L}\right) \sin\left(\frac{\pi y}{L}\right) \sin\left(\frac{2\pi z}{L}\right),$$

$$\psi_{2,2,1}(x,y,z) = \sqrt{\frac{8}{L^3}} \sin\left(\frac{2\pi x}{L}\right) \sin\left(\frac{2\pi y}{L}\right) \sin\left(\frac{\pi z}{L}\right).$$

- Third Excited State: For the third excited state, one typical set of quantum numbers is where one quantum number equals 3 and the other two equal 1:

$$\begin{cases} n_x = 1,\ n_y = 1,\ n_z = 3, \\ n_x = 1,\ n_y = 3,\ n_z = 1, \\ n_x = 3,\ n_y = 1,\ n_z = 1. \end{cases}$$

  The energy is

$$E_3 = \frac{\hbar^2\pi^2}{2mL^2}(1^2 + 1^2 + 3^2) = \frac{11\hbar^2\pi^2}{2mL^2},$$

  which again is threefold degenerate, with eigenfunctions:

$$\psi_{1,1,3}(x,y,z) = \sqrt{\frac{8}{L^3}} \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi y}{L}\right) \sin\left(\frac{3\pi z}{L}\right),$$

$$\psi_{1,3,1}(x,y,z) = \sqrt{\frac{8}{L^3}} \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{3\pi y}{L}\right) \sin\left(\frac{\pi z}{L}\right),$$

$$\psi_{3,1,1}(x,y,z) = \sqrt{\frac{8}{L^3}} \sin\left(\frac{3\pi x}{L}\right) \sin\left(\frac{\pi y}{L}\right) \sin\left(\frac{\pi z}{L}\right).$$

- Fourth Excited State: As a special case, consider the state with

$$n_x = n_y = n_z = 2.$$

  Its energy is

$$E_4 = \frac{\hbar^2\pi^2}{2mL^2}(2^2 + 2^2 + 2^2) = \frac{12\hbar^2\pi^2}{2mL^2},$$

  and the corresponding eigenfunction is

$$\psi_{2,2,2}(x,y,z) = \sqrt{\frac{8}{L^3}} \sin\left(\frac{2\pi x}{L}\right) \sin\left(\frac{2\pi y}{L}\right) \sin\left(\frac{2\pi z}{L}\right).$$

  This state is non-degenerate.

We observe that for the energy levels E_1, E_2, and E_3, there are three distinct eigenfunctions corresponding to the same energy eigenvalue—indicating a threefold degeneracy. This degeneracy arises from the symmetry of the cubic potential and the fact that different sets of quantum numbers (n_x, n_y, n_z) can end up with the same sum $n_x^2 + n_y^2 + n_z^2$. Such degeneracies reflect the symmetry and the completeness of the eigenbasis, which are central to the spectral theorem in linear algebra and quantum mechanics.

4.4 Hydrogen Atom

Another classical model of the application for Quantum Mechanics is solving the Hydrogen Atom, in this section, we will going through it together.

4.4.1 Schrödinger Equation in Spherical Coordinate

For an hydrogen atom, it is under 3-dimensional space, thus the potential of an electron can be represented as a form related to the distance to the nucleus.

$$U = U(\mathbf{x}) = U(r) \tag{4.4.1.1}$$

In such case, we are in a spherical coordinate, and have Schrödinger Equation

$$\left[-\frac{\hbar^2}{2\mu}\nabla^2 + U(r)\right]\psi = E\psi \tag{4.4.1.2}$$

where we have the Laplacian operator in spherical coordinate

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2} \tag{4.4.1.3}$$

Now, the Schrödinger Equation

$$-\frac{\hbar^2}{2\mu r^2}\left[\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)+\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]\psi + U(r)\psi = E\psi \tag{4.4.1.4}$$

rewrite into the form

$$\left[\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)+\frac{2\mu r^2}{\hbar^2}(E-U(r))\right]\psi = -\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right] \tag{4.4.1.5}$$

Now the left hand side and right hand side are separated, and we can set function

$$\psi(r,\theta,\varphi) = R(r)Y(\theta,\varphi) \tag{4.4.1.6}$$

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{2\mu r^2}{\hbar^2}[E-U(r)] = -\frac{1}{Y}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\varphi^2}\right] \tag{4.4.1.7}$$

Since the two side of this equation doesn't interrupt each other, we can write them separately, both equal to λ

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{2\mu r^2}{\hbar^2}[E-U(r)] = \lambda$$
$$-\frac{1}{Y}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\varphi^2}\right] = \lambda \tag{4.4.1.8}$$

4.4.2 Solving Spherical Harmonics

For the second equation, it is a classic form called spherical harmonics, it have no relation with potential.
Our strategy is still using seperation of variables

$$Y(\theta,\varphi) = \Theta(\theta)\Phi(\varphi) \tag{4.4.2.1}$$

substitute back

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Theta}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2\Phi}{\partial\varphi^2} = -\lambda\Theta\Phi \tag{4.4.2.2}$$

It can be again separate into two functions

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\left(\lambda-\frac{\nu}{\sin^2\theta}\right)\Theta = 0 \tag{4.4.2.3}$$

$$\frac{d^2\Phi}{d\varphi^2}+\nu\Phi = 0 \tag{4.4.2.4}$$

For (4.4.2.4) the solution is given by

$$\Phi(\varphi) = Ae^{i\sqrt{\nu}\varphi}+Be^{-i\sqrt{\nu}\varphi}, \quad \nu\neq 0 \tag{4.4.2.5}$$

$$\Phi(\varphi) = C+D\varphi, \quad \nu = 0 \tag{4.4.2.6}$$

For Φ, in physics, $\varphi+2\pi$ should have same value as φ, thus, $D=0$, and thus, $\sqrt{\nu}=0$, or integer m. Thus the solution is

$$\Phi_m(\varphi) = Ne^{im\varphi} \tag{4.4.2.7}$$

To find the normalization constant

$$\delta_{mn} = N^2\int_0^{2\pi} e^{im\varphi}e^{-im\varphi}\,d\varphi, \quad N=\frac{1}{\sqrt{2\pi}} \tag{4.4.2.8}$$

and substitute $\nu = m^2$ into (4.4.2.3)

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\left(\lambda-\frac{m^2}{\sin^2\theta}\right)\Theta = 0 \tag{4.4.2.9}$$

Also, substitute $\zeta=\cos\theta$, then

$$\frac{d\zeta}{d\theta} = -\sin\theta = -(1-\zeta)^{1/2}, \quad \frac{d}{d\theta} = -(1-\zeta)^{1/2}\frac{d}{d\zeta} \tag{4.4.2.10}$$

and

$$P(\zeta) = P(\cos\theta) = \Theta(\theta) \tag{4.4.2.11}$$

$$\frac{d}{d\zeta}\left[(1-\zeta^2)\frac{dP}{d\zeta}\right]+\left[\lambda-\frac{m^2}{1-\zeta^2}\right]P = 0 \tag{4.4.2.12}$$

change to the form

$$(1-\zeta^2)\frac{d^2P}{d\zeta^2}-2\zeta\frac{dP}{d\zeta}+\left[\lambda-\frac{m^2}{1-\zeta^2}\right]P = 0 \tag{4.4.2.13}$$

As shown in [5], this is the *A'ssociated Legendre Equation*, and the way to solve is with the form

$$P(\zeta) = (1-\zeta^2)^{|m|/2}\nu_m(\zeta) \tag{4.4.2.14}$$

Then this equation can be reduced to

$$(1-\zeta^2)\frac{d^2\nu_m}{d\zeta^2}-2(|m|+1)\zeta\frac{d\nu_m}{d\zeta}+(\lambda-|m|-m^2)\nu_m = 0 \tag{4.4.2.15}$$

This can be solved by power series

$$\nu_m(\zeta) = \sum_{\rho=0}^{\infty}a_\rho\zeta^\rho \tag{4.4.2.16}$$

substitute into (4.4.2.15), we have

$$\begin{aligned}
&(1-\zeta^2)\sum_{\rho=2}^{\infty}a_\rho(\rho-1)\rho\zeta^{\rho-2}-2(|m|+1)\zeta\sum_{\rho=1}^{\infty}a_\rho\rho\zeta^{\rho-1}\\
&+(\lambda-|m|-m^2)\sum_{\rho=0}^{\infty}a_\rho\zeta^\rho = 0\\
=&\sum_{\rho=2}^{\infty}a_\rho(\rho-1)\rho\zeta^{\rho-2}-\sum_{\rho=2}^{\infty}a_\rho(\rho-1)\rho\zeta^\rho\\
&-2(|m|+1)\sum_{\rho=1}^{\infty}a_\rho\rho\zeta^\rho+(\lambda-|m|-m^2)\sum_{\rho=0}^{\infty}a_\rho\zeta^\rho = 0\\
=&\sum_{\rho=0}^{\infty}a_{\rho+2}(\rho+1)(\rho+2)\zeta^\rho-\sum_{\rho=2}^{\infty}a_\rho(\rho-1)\rho\zeta^\rho\\
&-2(|m|+1)\sum_{\rho=1}^{\infty}a_\rho\rho\zeta^\rho+(\lambda-|m|-m^2)\sum_{\rho=0}^{\infty}a_\rho\zeta^\rho = 0
\end{aligned} \tag{4.4.2.17}$$

For $\rho\geq 2$

$$a_{\rho+2}(\rho+1)(\rho+2)-[(\rho-1)\rho-2(|m|+1)\rho+(\lambda-|m|-m^2)]a_\rho = 0 \tag{4.4.2.18}$$

As we can see the coefficient of expansion, if ν is infinite series, it must diverge, but since physics require us to have finite function, thus ν should have finite items. Set $\rho=k$ as the highest power of this polynomials, then, $a_{k+2}=0$, given

$$\begin{aligned}
&k(k-1)+2(|m|+1)k-\lambda+|m|+m^2 = 0\\
&\lambda = (k+|m|)(k+|m|+1) = l(l+1)\\
&l = k+|m|
\end{aligned} \tag{4.4.2.19}$$

Here, after later experiment and observation, we know that l is orbital quantum number, it is clearly non-negative integer number. After l is determined, $|m|$ can not be value larger then l, and it is called magnetic quantum number. A later conclusion is that l can represent the magnitude of angular momentum on certain orbit, and m represents the magnitude the angular momentum projected on some axle.
We set

$$\begin{aligned}&\lambda = l(l+1),\\ &l = 0, 1, 2, 3, \cdots\\ &m = 0, \pm 1, \pm 2, \cdots, \pm l\end{aligned} \tag{4.4.2.20}$$

Then the equation (4.4.2.13) can be written as

$$(1-\zeta^2)\frac{d^2P}{d\zeta^2} - 2\zeta\frac{dP}{d\zeta} + \left[l(l+1) - \frac{m^2}{1-\zeta^2}\right]P = 0 \tag{4.4.2.21}$$

we can first discuss the case where $m = 0$, the solution denoted P_l

$$(1-\zeta^2)\frac{d^2P_l}{d\zeta^2} - 2\zeta\frac{dP_l}{d\zeta} + l(l+1)P_l = 0 \tag{4.4.2.22}$$

By (4.4.2.18) and (4.4.2.16)

$$P_l(\zeta) = \sum_{\rho=0}^{\infty} a_\rho \zeta^\rho \tag{4.4.2.23}$$

$$a_{\rho+2} = \frac{\rho(\rho+1) - l(l+1)}{(\rho+2)(\rho+1)}a_\rho \tag{4.4.2.24}$$

Since this recursion relation only related the coefficient in even or odd index, we can obtain both of them by setting $a_0 \neq 0, a_1 = 0$ and $a_0 = 0, a_1 \neq 0$. And to satisfy this polynomial, we also set

$$P_l(1) = 1 \tag{4.4.2.25}$$

Then, this polynomial can be represented by

$$P_l(\zeta) = \frac{1}{2^l l!}\frac{d^l}{d\zeta^l}(\zeta^2-1)^l \tag{4.4.2.26}$$

it can be shown that this equation satisfies (4.4.2.22)
For the case $\nu \neq 0$, if we differentiate (4.4.2.15),

$$\begin{aligned}&(1-\zeta^2)\nu_m''' - 2\zeta v_m'' - 2(|m|+1)\zeta\nu_m'' - 2(|m|+1)\nu_m'\\ &+ (\lambda - |m| - m^2)\nu_m' = 0\\ =&(1-\zeta^2)\nu_m''' - 2[(|m|+1)+1]\zeta\nu_m''\\ &+ [\lambda - (|m|+1) - (|m|+1)^2]\nu_m' = 0\end{aligned} \tag{4.4.2.27}$$

comparing with (4.4.2.15), this is the equation $\nu_{|m|+1}$ satisfies, thus the difference is just a coefficient.

$$\begin{aligned}&\nu_0 = P_l(\zeta)\\ &v_{|m|+1}(\zeta) = C\frac{d^{|m|}}{d\zeta^{|m|}}P_l(\zeta)\end{aligned} \tag{4.4.2.28}$$

thus, we get the solution to (4.4.2.21)

$$P_l^{|m|}(\zeta) = (-1)^{|m|}(1-\zeta^2)^{\frac{|m|}{2}}\frac{d^{|m|}}{d\zeta^{|m|}}P_l(\zeta) \tag{4.4.2.29}$$

it is the *Associated Legendre Polynomials*.
There is some symmetry of this polynomials

$$P_l^{|m|}(-\zeta) = (-1)^{l+|m|}P_l^{|m|}(\zeta) = (-1)^{l+m}P_l^{|m|}(\zeta) \tag{4.4.2.30}$$

for convinience, we define

$$P_l^{-m}(\zeta) = (-1)^m\frac{(l-m)!}{(l+m)!}P_l^m(\zeta), \quad m > 0 \tag{4.4.2.31}$$

And thus, we conclude with a function

$$Y_{lm}(\theta,\varphi) = N_{lm}P_l^m(\cos\theta)e^{im\varphi} \tag{4.4.2.32}$$

This is called *Spherical Harmonic Function* and N_{lm} is the normalization constant, it is given by

$$\int_0^{2\pi}\int_0^{\pi} Y_{lm}^* Y_{l'm'}\sin\theta\, d\theta\, d\varphi = \delta_{ll'}\delta_{mm'} \tag{4.4.2.33}$$

$$N_{lm} = \sqrt{\frac{(2l+1)(l-m)!}{4\pi(l+m)!}} \tag{4.4.2.34}$$

Below, given some spherical harmonics

$$\begin{aligned}&Y_{00} = \frac{1}{\sqrt{4\pi}}\\ &Y_{11} = -\sqrt{\frac{3}{8\pi}}\sin\theta e^{i\varphi}, \quad Y_{10} = -\sqrt{\frac{3}{4\pi}}\cos\theta,\\ &Y_{1,-1} = \sqrt{\frac{3}{8\pi}}\sin\theta e^{-i\varphi}\\ &Y_{22} = \sqrt{\frac{15}{32\pi}}\sin^2\theta e^{i2\varphi}, \quad Y_{21} = -\sqrt{\frac{15}{8\pi}}\sin\theta\cos\theta e^{i\varphi}\\ &Y_{20} = \sqrt{\frac{5}{16\pi}}(3\cos^2\theta - 1)\\ &Y_{2,-1} = \sqrt{\frac{15}{8\pi}}\sin\theta\cos\theta e^{-i\varphi}, \quad Y_{2,-2} = \sqrt{\frac{15}{32\pi}}\sin^2\theta e^{-i2\varphi}\end{aligned} \tag{4.4.2.35}$$

Traditionally, we say the wave function represented by orbit quantum number $l = 0, 1, 2, 3, \cdots$ as s, p, d, f, $\cdots$ states.

4.4.3 Radial Wave Function

Now, let's deal with the first equation in (4.4.1.8). Substitute $R(r)$ and $\lambda = l(l+1)$ into the equation.

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \left\{\frac{2\mu}{\hbar^2}[E - U(r)] - \frac{l(l+1)}{r^2}\right\}R = 0 \tag{4.4.3.1}$$

Since a singularity is $r = 0$, a trick is to substitute $R(r) = \frac{u(r)}{r}$ into the equation.

$$u'' + \left\{\frac{2\mu}{\hbar^2}[E - U(r)] - \frac{l(l+1)}{r^2}\right\}u = 0 \tag{4.4.3.2}$$

which is equal to be thinking as a particle in half 1-dimension with a potential $U(r) = \frac{\hbar^2}{2\mu}\frac{l(l+1)}{r^2}$ from $0 \to \infty$. Under this potential, the item related to l is positive, meaning that it is the repel potential, higher angular momentum, the higher repel. Set the form of $U(r)$

$$U(r) = A/r^s, \quad s > 0 \tag{4.4.3.3}$$

When $A > 0$, it is repelling potential, and when $A < 0$, it is attracting potential.

$$u'' + \left\{\frac{2\mu}{\hbar^2}E - \frac{2\mu A}{\hbar^2 r^2} - \frac{l(l+1)}{r^2}\right\}u = 0 \tag{4.4.3.4}$$

for this differential equation, we can analysis the condition where $r \to \infty$,

$$U(r) \to 0, \quad \frac{1}{r^2} \to 0 \tag{4.4.3.5}$$

In this case, we have the asymptote equation

$$u'' + \frac{2\mu E}{\hbar^2} u = 0 \tag{4.4.3.6}$$

For this equation, we are familiar with, which has solution

$$u(r) = C_1 e^{ikr} + C_2 e^{-ikr}, \quad k = \frac{\sqrt{2\mu E}}{\hbar} \tag{4.4.3.7}$$

$$R(r) = \frac{u(r)}{r} = \frac{C_1}{r} e^{ikr} + \frac{C_2}{r} e^{-ikr}, \quad r \to \infty \tag{4.4.3.8}$$

As whatever value E is, this asymptotic R satisfies the condition of a wave function. After experiment and realization later, the two terms, first is called *Outgoing Spherical Wave*, and second is called *Ingoing Spherical Wave*. And the first term is representing the particle moving from center to infinite, and the second is representing particle moving towards center, it is equal to the non-periodic orbital motion in classical mechanics, which is scattering problem in quantum mechanics. But we will not approaching there in this article.

And for the case $E < 0$, where only attracting potential affects, since the second term diverges, does not fulfill the requirement for a wave function, thus we only have

$$R(r) = B_1 e^{-\beta r}/r, \quad \beta = \sqrt{2\mu(-E)/\hbar^2} \tag{4.4.3.9}$$

In this case, it is easy to see

$$|R(r)|^2 = |B_1|^2 \frac{e^{-2\beta r}}{r^2} \to 0, \quad r \to \infty \tag{4.4.3.10}$$

meaning that for the position nearing infinity, the probability for particle to appear in infinitely small, similar to our intuition.
And let's observe how $u(r)$ behave when $r \to 0$

Let's first set $s < 2$, when $r \to 0$, after comparing the speed of divergence between A/r^s and $l(l+1)/r^2$, the r^{-2} diverges stronger then $r^{-s}, s < 2$ as $r \to 0$. Thus A/r^s can be omitted, and we have the approximated equation

$$u'' - \frac{l(l+1)}{r^2} u = 0, \quad r \to 0 \tag{4.4.3.11}$$

By [6], we can see this equation has regular singularity, which can use *Laurent Expansions*, we set

$$u(r) = r^\rho \sum_{\nu=0}^{\infty} b_\nu r^\nu = \sum_{\nu=0}^{\infty} b_\nu r^{\nu+\rho} \tag{4.4.3.12}$$

plug into the equation (4.4.3.11), we have

$$\begin{aligned} u' &= \sum_{\nu=1}^{\infty} b_\nu (\nu+\rho) r^{\nu+\rho-1} \\ u'' &= \sum_{\nu=2}^{\infty} b_\nu (\nu+\rho-1)(\nu+\rho) r^{\nu+\rho-2} \\ &\sum_{\nu=2}^{\infty} [b_\nu (\nu+\rho-1)(\nu+\rho) - (l(l+1))b_\nu] r^{\nu+\rho-2} = 0 \end{aligned} \tag{4.4.3.13}$$

when expending with r, where $\nu = 0$

$$[(\rho-1)\rho - l(l+1)] = 0 \tag{4.4.3.14}$$

Thus

$$\begin{cases} \rho = -l \\ \rho = l+1 \end{cases} \tag{4.4.3.15}$$

in case $\rho = -l$ have

$$\begin{aligned} u(r) = r^{-l} \sum b_\nu r^\nu \to \infty, \quad r \to 0 \\ R(r) = u(r)/r \to \infty, \quad r \to 0 \end{aligned} \tag{4.4.3.16}$$

is not the nature the wave function behaves, thus we take $\rho = l+1$

$$\begin{aligned} u(r) = r^{l+1} \sum b_\nu r^\nu \to 0, \quad r \to 0 \\ R(r) = u(r)/r \to 0, \quad r \to 0 \end{aligned} \tag{4.4.3.17}$$

Satisfies the condition for wave function. The larger l, the faster R approaching 0, meaning that harder for particle to going through the center of the atom.

Now, let's look at the case $s > 2$, in this case, since A/r^s diverges faster then r^{-2}, thus

$$u'' - \frac{2\mu}{\hbar^2} \frac{A}{r^s} u = 0 \tag{4.4.3.18}$$

similarly with before, take

$$u(r) = r^\rho \sum_{\nu=0}^{\infty} b_\nu r^\nu = \sum_{\nu=0}^{\infty} b_\nu r^{\nu+\rho} \tag{4.4.3.19}$$

Then

$$\sum_{\nu=2}^{\infty} [(\nu+\rho-1)(\nu+\rho) - \frac{2\mu A}{\hbar^2}] b_\nu r^{\nu+\rho-2} = 0 \tag{4.4.3.20}$$

For this, since $s > 2$, the coefficient $-\frac{2\mu A b_0}{\hbar^2}$ of the lowest power of $r^{\rho-\nu}$ must be 0. Thus $A = 0$, which we can infer to potential equals to 0, which is impossible. Besides, in fact, when $s > 2$, $r = 0$ is irregular singularity, thus does not exist regular solution. Thus, there are no solution for $s > 2$.

In conclusion, when the particle have energy greater then zero, meaning it is in the outgoing state; when the particle have energy lower then zero, the existence of bound state only when it is in attracting potential with $s < 2$

4.4.4 Coulomb Potential

By definition of electric potential

$$u(r) = -Q/r, \quad Q > 0 \tag{4.4.4.1}$$

Thus, if there is a charge with $-e$ under the field of a particle with charge Ze. $Q = Ze^2/4\pi\varepsilon_0$
Under this potential, there exist bound state. In this case, $E < 0$, thus we have radial wave function

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{dR}{dr}\right) + \left\{\frac{2\mu}{\hbar^2}\left[E + \frac{Q}{r}\right] - \frac{l(l+1)}{r^2}\right\} R = 0 \tag{4.4.4.2}$$

Set $R = u/r$

$$u'' + \left[\frac{2\mu}{\hbar^2}\left(E + \frac{Q}{r}\right) - \frac{l(l+1)}{r^2}\right] u = 0 \tag{4.4.4.3}$$

For simplicity, we define

$$\alpha = \left(\frac{8\mu|E|}{\hbar^2}\right)^{1/2}, \quad \beta = \frac{2\mu Q}{\alpha\hbar^2} = \frac{Q}{\hbar}\left(\frac{\mu}{2|E|}\right)^{1/2} \tag{4.4.4.4}$$

and do substitution

$$\rho = \alpha r, \quad u(r) = u\left(\frac{\rho}{\alpha}\right) \equiv v(\rho) \tag{4.4.4.5}$$

simplify the equation to

$$\frac{d^2v}{d\rho^2} + \left[\frac{\beta}{\rho} - \frac{1}{4} - \frac{l(l+1)}{\rho^2}\right] v = 0 \tag{4.4.4.6}$$

When $\rho \to \infty$, the asymptotic equation is

$$\frac{d^2v}{d\rho^2} - \frac{1}{4}v = 0 \quad (\rho \to \infty) \tag{4.4.4.7}$$

Then, the solution is given by

$$v(\rho) = e^{\pm\rho/2} \quad (\rho \to \infty) \tag{4.4.4.8}$$

And since only $e^{-\rho/2}$ satisfies the condition for wave function, with the consideration of this asymptotic property, we can let

$$v(\rho) = e^{-\rho/2} f(\rho) \tag{4.4.4.9}$$

When plug into the (4.4.4.7),

$$\frac{d^2f}{d\rho^2} - \frac{df}{d\rho} + \left[\frac{\beta}{\rho} - \frac{l(l+1)}{\rho^2}\right] f = 0 \tag{4.4.4.10}$$

we can still use power series to solve this equation

$$\begin{aligned} f &= \sum_{\nu=0}^{\infty} b_\nu \rho^{\nu+\gamma} \\ f' &= \sum_{\nu=1}^{\infty} b_\nu (\nu+\gamma)\rho^{\nu+\gamma-1} \\ f'' &= \sum_{\nu=2}^{\infty} b_\nu (\nu+\gamma)(\nu+\gamma-1)\rho^{\nu+\gamma-2} \end{aligned} \tag{4.4.4.11}$$

Then

$$\begin{aligned} &\sum_{\nu=2}^{\infty} b_\nu(\nu+\gamma)(\nu+\gamma-1)\rho^{\nu+\gamma-2} - \sum_{\nu=1}^{\infty} b_\nu(\nu+\gamma)\rho^{\nu+\gamma-1} \\ &+ \left[\frac{\beta}{\rho} - \frac{l(l+1)}{\rho^2}\right]\sum_{\nu=0}^{\infty} b_\nu \rho^{\nu+\gamma} = 0 \\ &\sum_{\nu=0}^{\infty}[[(\nu+\gamma+1)(\nu+\gamma)b_{\nu+2} - (\nu+\gamma+1)b_{\nu+1}]\rho^{\nu+\gamma} \\ &+ \beta b_{\nu+1}\rho^{\nu+\gamma-1} - l(l+1)b_{\nu+2}\rho^{\nu+\gamma-2}] = 0 \end{aligned} \tag{4.4.4.12}$$

compare the terms with same power,

$$\begin{aligned} &[(\nu+\gamma)(\nu+\gamma+1) - l(l+1)]\, b_{\nu+1} - (\nu+\gamma+\beta)b_\nu = 0 \\ &b_{\nu+1} = \frac{-\beta + (\nu+\gamma)}{(\nu+\gamma)(\nu+\gamma+1) - l(l+1)} b_\nu \end{aligned} \tag{4.4.4.13}$$

by this relation, as $\nu \to \infty$, must have

$$\frac{b_{\nu+1}}{b_\nu} = \frac{1}{v} \tag{4.4.4.14}$$

Thus, f has same coefficient relationship with

$$e^\rho = \sum \frac{\rho^\nu}{\nu!} \tag{4.4.4.15}$$

and thus, v is similar with $e^{\rho/2}$ which is not the condition for it. Thus, f must be a polynomial with finite terms, suppose the highest order is

$$b_{n_r}\rho^{n_r+\gamma}, \quad b_{n_r+1} = 0 \tag{4.4.4.16}$$

and thus

$$0 = \frac{-\beta + (n_r+\gamma)}{(n_r+\gamma)(n_r+\gamma+1) - l(l+1)}, \quad \beta = n_r + \gamma \tag{4.4.4.17}$$

and since this series begins at ρ^γ, thus

$$b_{-1} = \frac{(-1+\gamma)(-1+\gamma+1) - l(l+1)}{-\beta - 1 + \gamma} b_0 \tag{4.4.4.18}$$

thus

$$(\gamma-1)\gamma - l(l+1) = 0, \quad \begin{cases} \gamma = -l \\ \gamma = l+1 \end{cases} \tag{4.4.4.19}$$

since when $\gamma = -l$, it diverges when $r \to 0$, thus does not align with physics, thus we can only take $\gamma = l+1$. Thus we have

$$\beta = n_r + l + 1 = n \tag{4.4.4.20}$$

in which n_r is called radial quantum number, and n is principal quantum number. Since n_r, l are non-negative number, thus $n = 1, 2, 3, \cdots$ are integer number. Substitute back into (4.4.4.4), we have

$$E_n = -\frac{\mu Q^2}{2\hbar^2 n^2}, \quad n = 1, 2, 3, \cdots \tag{4.4.4.21}$$

meaning that, under bound state, the energy of particle are in discrete number, which forming the energy level.
Substitute back β, γ into (4.4.4.13), we have

$$\begin{aligned} b_{\nu+1} &= -\frac{n-l-1}{(\nu+1)(2l+2+\nu)} b_\nu \\ f(\rho) &= b_0\rho^{l+1}\left[1 - \frac{n-l-1}{1!(2l+2)}\rho + \cdots\right] \\ &= -b_0 \frac{(2l+1)!(n-l-1)!}{[(n+l)!]^2}\rho^{l+1} L_{n+l}^{2l+1}(\rho) \end{aligned} \tag{4.4.4.22}$$

in which

$$L_{n+l}^{2l+1}(\rho) = \sum_{\nu=0}^{n-l-1} (-1)^{\nu+1} \frac{[(n+l)!]^2\rho^\nu}{(n-l-1-\nu)!(2l+1+\nu)!\nu!} \tag{4.4.4.23}$$

called *Associated Laguerre Polynomials*, one can be also represented as

$$L_{n+l}^{2l+1}(\rho) = \frac{d^{2l+1}}{d\rho^{2l+1}}\left[e^\rho \frac{d^{n+l}}{d\rho^{n+l}}(e^{-\rho}\rho^{n+l})\right] = \frac{d^{2l+1}}{d\rho^{2l+1}} L_{n+1}(\rho) \tag{4.4.4.24}$$

and

$$L_{n+1}(\rho) = e^\rho \frac{d^{n+l}}{d\rho^{n+l}}(e^{-\rho}\rho^{n+l}) \tag{4.4.4.25}$$

called *Laguerre Polynomials*

Now, we have

$$\alpha = \alpha_n = \left(\frac{8\mu|E_n|}{\hbar^2}\right)^{1/2} = \frac{2\mu Q}{\hbar^2 n}, \quad \rho = \alpha_n r \tag{4.4.4.26}$$

Thus, now, the radial wave function can be written as

$$\begin{aligned} R_{nl}(r) &= \frac{1}{2}u(r) = \frac{1}{r}v(\rho) = \frac{1}{\alpha_n r}N_{nl}e^{-\rho/2}f(\rho) \\ &= N_{nl}\frac{1}{\alpha_n r}e^{-\frac{1}{2}\alpha_n r}f(\alpha_n r) \\ &= N_{nl}\frac{1}{\alpha_n r}e^{-\frac{1}{2}\alpha_n r}L_{n+l}^{2l+1}(\alpha_n r) \\ &= N_{nl}e^{-\frac{\mu Q}{\hbar^2 n}}\left(\frac{2\mu Q}{\hbar^2 n}\right)^l L_{n+l}^{2l+1}\left(\frac{2\mu Q}{\hbar^2 n}r\right) \end{aligned} \tag{4.4.4.27}$$

in which N_{nl} is normalization constant,

$$\begin{aligned} 1 &= \int_0^{2\pi}\int_0^{\pi}\int_0^{\infty}\psi^*\psi\, r^2\sin\theta\, dr d\theta d\varphi \\ &= \int_0^{\infty}|R_{nl}(r)|^2 r^2 dr \iint Y_{lm}^* Y_{lm}\sin\theta d\theta d\varphi \\ &= \int_0^{\infty}|R_{nl}(r)|^2 r^r dr \end{aligned} \tag{4.4.4.28}$$

Then

$$\begin{aligned} N_{nl} &= -\left\{\alpha_n^3\frac{(n-l-1)!}{2n[(n+l)!]^3}\right\}^{\frac{1}{2}} \\ &= -\left\{\left(\frac{2\mu Q}{\hbar^2 n}\right)^3\frac{(n-l-1)!}{2n[(n+l)!]^3}\right\}^{\frac{1}{2}} \end{aligned} \tag{4.4.4.29}$$

below is some radial wave function

$$\begin{aligned} R_{10}(r) &= \left(\frac{\mu Q}{\hbar^2}\right)^{\frac{3}{2}}2e^{-\frac{\mu Q}{\hbar^2}r} \\ R_{21}(r) &= \left(\frac{\mu Q}{\hbar^2}\right)^{\frac{3}{2}}\frac{\mu Q}{3\hbar^2}re^{-\frac{\mu Q}{\hbar^2}r} \\ R_{20}(r) &= \left(\frac{\mu Q}{\hbar^2}\right)^{\frac{3}{2}}\left(2-\frac{\mu Q}{\hbar^2}r\right)e^{-\frac{\mu Q}{\hbar^2}r} \end{aligned} \tag{4.4.4.30}$$

Let's have a small summary,
Under the potential of

$$U(r) = -Q/r, \quad Q > 0$$

the particle have bound state, with wave function

$$\psi_{nlm}(r,\theta,\varphi) = R_{nl}(r)Y_{lm}(\theta,\varphi) \tag{4.4.4.31}$$

where R and Y can be seen in (4.4.4.30) and (4.4.2.35), with energy level

$$E_n = -\frac{\mu Q}{2\hbar^2}\frac{1}{n^2}, \quad n = 1,2,3,\cdots \tag{4.4.4.32}$$

then, when principal quantum number is confirmed, the energy of particle is confirmed, but the wave function can not confirmed in this case since l can take

$$l = 0,1,2,3,\cdots,n-1 \tag{4.4.4.33}$$

with different l, m, the wave function is different, which have total

$$\sum_{l=0}^{n-1}\sum_{m=-l}^{l}1 = \sum_{l=0}^{n-1}(2l+1) = n^2 \tag{4.4.4.34}$$

meaning that for a energy level with index n, there are total n^2 different wave function, we call this n^2 fold degeneracy.

4.4.5 Hydrogen Atom

After preparation of Spherical Harmonics, Radial Wave Function and many analysis on their properties, we finally step to the main problem we want to solve in this section - Hydrogen Atom.

Hydrogen Atom is a stable system with one electron and one nucleus and one proton. which is a two particle system, and we can construct the equation of motion.

We set the position of electron as $\mathbf{x}_e$ and position of nucleus as $\mathbf{x}_N$, then the position of electron with respect to nucleus is given by

$$\mathbf{x} = \mathbf{x_e} - \mathbf{x}_N \tag{4.4.5.1}$$

The total energy of hydrogen atom equals to the kinetic energy of electron and proton, and the potential energy in between, which can be represented as

$$E_T = \frac{\mathbf{p}_e^2}{2m_e} + \frac{\mathbf{p}_N^2}{2m_N} + U(|\mathbf{x}_e - \mathbf{x}_N|) \tag{4.4.5.2}$$

and can represent with operator

$$\begin{aligned} &E_T\Psi(\mathbf{x}_e,\mathbf{x}_N) \\ &= \left[-\frac{\hbar^2\nabla_{\mathbf{x}_e}^2}{2m_e} - \frac{\hbar^2\nabla_{\mathbf{x}_N}^2}{2m_N} + U(|\mathbf{x}_e - \mathbf{x_N}|)\right]\Psi(\mathbf{x}_e,\mathbf{x_N}) \end{aligned} \tag{4.4.5.3}$$

changing to the barycentric coordinate and relative coordinate

$$\begin{aligned} \frac{\partial}{\partial x_{ei}} &= \sum_{j=1}^{3}\left(\frac{\partial X_j}{\partial x_{ei}}\frac{\partial}{\partial X_j} + \frac{\partial x_j}{\partial x_{ei}}\frac{\partial}{\partial x_j}\right) = \frac{m_e}{M}\frac{\partial}{\partial X_i} + \frac{\partial}{\partial x_i} \\ \frac{\partial}{\partial X_{Ni}} &= \frac{m_N}{M}\frac{\partial}{\partial X_i} - \frac{\partial}{\partial x_i} \end{aligned} \tag{4.4.5.4}$$

similarly

$$\begin{aligned} \nabla_{\mathbf{x}_e} &= \frac{m_e}{M}\nabla_{\mathbf{X}} + \nabla_{\mathbf{x}} \\ \nabla_{\mathbf{x}_N} &= \frac{m_N}{M}\nabla_{\mathbf{X}} - \nabla_{\mathbf{x}} \end{aligned} \tag{4.4.5.5}$$

and

$$\begin{aligned} \nabla_{x_e}^2 &= \frac{m_e^2}{M^2}\nabla_X^2 + 2\frac{m_e}{M}(\nabla_{\mathbf{X}}\cdot\nabla_{\mathbf{x}}) + \nabla_x^2 \\ \nabla_{x_N}^2 &= \frac{m_N^2}{M^2}\nabla_X^2 + 2\frac{m_N}{M}(\nabla_{\mathbf{X}}\cdot\nabla_{\mathbf{x}}) + \nabla_x^2 \end{aligned} \tag{4.4.5.6}$$

$$\begin{aligned} \frac{\nabla_{\mathbf{x}_e}^2}{2m_e} + \frac{\nabla_{\mathbf{x}_N}^2}{2m_N} &= \frac{1}{2M}\nabla_X^2 + \frac{1}{2}\left(\frac{1}{m_e} + \frac{1}{m_N}\right)\nabla_x^2 \\ &= \frac{1}{2M}\nabla_X^2 + \frac{1}{2\mu}\nabla_X^2 + \frac{1}{2\mu}\nabla_x^2 \end{aligned} \tag{4.4.5.7}$$

here, $\mu = \dfrac{m_e m_N}{m_e + m_n} = \dfrac{m_e m_N}{M}$ is called *Reduced atomic mass.* Then

$$E_T\Psi(\mathbf{X},\mathbf{x}) = \left[-\frac{\hbar^2\nabla_X^2}{2m_e} - \frac{\hbar^2\nabla_x^2}{2m_N} + U(|\mathbf{x}|)\right]\Psi(\mathbf{X},\mathbf{x}) \tag{4.4.5.8}$$

one can be solving with separation of variables. Let

$$\Psi(\mathbf{X},\mathbf{x}) = \psi(\mathbf{x})\phi(\mathbf{X}) \tag{4.4.5.9}$$

and each

$$-\frac{\hbar^2}{2\mu}\nabla_x^2\psi(\mathbf{x}) + U(|\mathbf{x}|)\psi(\mathbf{x}) = E\psi(\mathbf{x}) \tag{4.4.5.10}$$

$$-\frac{\hbar^2}{2M}\nabla_X^2\phi(\mathbf{X}) = (E_T - E)\phi(\mathbf{X}) \tag{4.4.5.11}$$

In Equation (4.4.5.10), the wave function $\psi(\mathbf{x})$ describes the relative motion of the electron with respect to the nucleus, where μ is the reduced mass and $U(|\mathbf{x}|)$ is the interaction potential between them. Equation (4.4.5.11) describes the motion of the center of mass of the hydrogen atom, with total mass M and $\phi(\mathbf{X})$ representing the center-of-mass wave function; here, E_T is the total energy and E is the energy associated with the internal motion.

This separation shows that the motion of a hydrogen atom can be decoupled into the relative (internal) motion and the overall translational (center-of-mass) motion.

Thus, the relative motion of the electron is equivalent to that of a particle with reduced mass μ moving in a central potential $U(r)$. Consequently, we can apply the results discussed previously. For the hydrogen atom, the Coulomb potential is given by

$$U(r) = -\frac{e^2}{4\pi\varepsilon_0 r},$$

and for a hydrogen-like atom (with a nucleus of charge $+Ze$), the potential becomes

$$U(r) = -\frac{Ze^2}{4\pi\varepsilon_0 r}.$$

By applying the solution we obtained earlier for such a central potential, the energy levels are

$$E_n = -\frac{\mu e^4}{2(4\pi\varepsilon_0)^2\hbar^2 n^2}, \quad n = 1, 2, 3, \ldots$$

The energy required to completely ionize the hydrogen atom (i.e., to separate the electron from the nucleus) is known as the *Ionization Energy*. For the ground state of hydrogen, this is

$$E_{\text{Ionization}} = 0 - E_1 = -E_1 = 13.6\,\text{eV}.$$

And the energy levels of hydrogen atom is given by Figure 5.

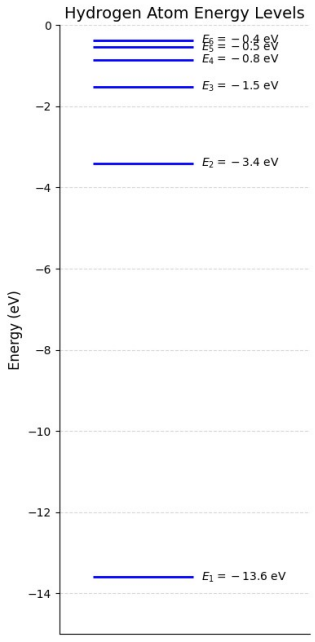


Figure 5: The Energy Level for Hydrogen

5 Perturbation Theorem

A *Perturbation* is meant to be a tiny interruption to the Hamiltonian of a quantum system, for example, a small protrusion in a one-dimensional infinite square wall. Generally, we suppose the Hamiltonian can be written as

$$H = H_0 + H' \tag{5.0.0.1}$$

in which the H_0 denoted the Hamiltonian without perturbation and H' denoted the perturbation Hamiltonian. Here, H_0 can be solved accurately

$$H_0\,|\phi_n^0\rangle = E_n^0\,|\phi_n^0\rangle \tag{5.0.0.2}$$

here, E_n^0 is the eigenenergy with eigenstate $|\phi_n^0\rangle$ corresponded, they fulfill the orthogonality and completeness

$$\begin{aligned}\langle\phi_n^0|\phi_m^0\rangle &= \delta_{nm}\\ \sum_n |\phi_n^0\rangle\,\langle\phi_n^0| &= 1\end{aligned} \tag{5.0.0.3}$$

let H' to be small, which is the perturbation on H_0, naming *Perturbation Operator*, thus, we can written like

$$H' = \varepsilon W \tag{5.0.0.4}$$

here, ε is a small real parameter, naming *Perturbation Parameter*, and W is the *Effective Perturbation Operator*.
And since the existence of perturbation, the eigen-equation of the system is now

$$H\,|\Psi\rangle = E\,|\Psi\rangle \tag{5.0.0.5}$$

and normally, we cannot solve it accurately, and to solve it, we need to use the accurate solution of H_0 to find a approximate solution, and this is called *Perturbation Theorem*.

5.1 Steady State Perturbation Theorem

Let the steady state Hamiltonian of the system to be $H = H_0 + H'$, the eigen-equation is

$$(H_0 + H')\,|\Psi\rangle = E\,|\Psi\rangle \tag{5.1.0.1}$$

let $|\Psi\rangle$ can be expanded with eigenfunction of H_0

$$|\Psi\rangle = \sum_n a_n\,|\phi_n^0\rangle \tag{5.1.0.2}$$

substitute into (5.1.0.1) we have

$$(H_0 + H')\sum_n a_n\,|\phi_n^0\rangle = E\sum_n a_n\,|\phi_n^0\rangle \tag{5.1.0.3}$$

do projection on $\langle\phi_m^0|$ we have

$$\begin{aligned}0 &= \langle\phi_m^0|\sum_n a_n(E_n^0 - E + H')\,|\phi_n^0\rangle\\ &= \sum_n a_n[(E_n^0 - E)\,\langle\phi_m^0|\phi_n^0\rangle] + \sum_n a_n\,\langle\phi_m^0|H'|\phi_n^0\rangle\\ &= a_m(E_m^0 - E) + \sum_n a_n\,\langle\phi_m^0|H'|\phi_n^0\rangle\end{aligned} \tag{5.1.0.4}$$

when we move out the term with $m = n$ we obtain

$$a_m(E_m^0 - E + H'_{mm}) + \sum_{n(n\neq m)} a_n H'_{mn} = 0 \tag{5.1.0.5}$$

here,

$$H'_{mn} = \langle \phi^0_m | H' | \phi^0_n \rangle \tag{5.1.0.6}$$

and it is called the *perturbation matrix elements*
Since the perturbation $H' = \varepsilon W$ is small, we can do expansion for a_m and eigenenergy E:

$$\begin{aligned} a_m &= a_m^{(0)} + \varepsilon a_m^{(1)} + \varepsilon^2 a_m^{(2)} + \cdots \\ E &= E^{(0)} + \varepsilon E^{(1)} + \varepsilon^2 E^{(2)} + \cdots \end{aligned} \tag{5.1.0.7}$$

here, the number in the parentheses is the order of approximation, for example $a_m^{(2)}$ is the expansion coefficient in second order approximation, and $\varepsilon E^{(1)}$ is the *energy correction* in first order. Now, we substitute (5.1.0.7) back into (5.1.0.5)

$$\begin{aligned} &(E_m^0 - E^{(0)})a_m^{(0)} \\ &+ \varepsilon[(W_{mm} - E^{(1)})a_m^{(0)} + (E_m^0 - E^{(0)})a_m^{(1)} + \sum_{n(n\neq m)} W_{mn}a_n^{(0)}] \\ &+ \varepsilon^2[(W_{mm} - E^{(1)})a_m^{(1)} + (E_m^0 - E^{(0)})a_m^{(2)} - E^{(2)}a_m^{(0)} \\ &+ \sum_{n(n\neq m)} W_{mn}a_n^{(1)}] \\ &+ \varepsilon^3[\cdots] \\ &+ \cdots \\ &= 0 \end{aligned} \tag{5.1.0.8}$$

(5.1.0.8) is called *Perturbation Theory Equation*, with this equation, we can find the eigenenergy for different approximation order, for example, $\varepsilon = 0$ (without perturbation) we have

$$(E_m^0 - E^{(0)})a_m^{(0)} = 0 \tag{5.1.0.9}$$

here, m is index for k^{th} eigenstate of H_0, thus, here we have

$$E^{(0)} = E_k^0, \quad a_m^{(0)} = \delta_{mk} \tag{5.1.0.10}$$

and this is the result of 0th order approximation relative to H_0, the eigenvalue is E_k^0 and eigenfunction is $|\phi_k^0\rangle$

5.1.1 First Order Approximation for Energy and Wavefunctions

substitute(5.1.0.10) into the perturbation theorem equation and take the first order term for ε, we have

$$\begin{aligned} &(E_m^0 - E_k^0)\delta_{mk} \\ &+ \varepsilon[(W_{mm} - E^{(1)})\delta_{mk} + (E_m^0 - E_k^0)a_m^{(1)} + \sum_{n(n\neq m)} W_{mn}\delta_{nk}] \\ &= 0 \end{aligned} \tag{5.1.1.1}$$

here, the first term is zero (no matter m, k). And for the case $m = k$ the second term in the square bracket is also zero, and the third term require $n \neq k$ to do the summation, thus, only remains $(W_{mm} - E^{(1)})\delta_{mk} = 0$, given that $E^{(1)} = W_{kk}$. Thus, the first order correction for the energy is

$$\varepsilon E^{(1)} = \varepsilon W_{kk} = H'_{kk} = \langle \phi_k^0 | H' | \phi_k^0 \rangle \tag{5.1.1.2}$$

this is the expectation value of the perturbation operator under non-perturbation state, thus, the first order approximation of the system energy is

$$E = E_k^0 + H'_{kk} = E_k^0 + \langle \phi_k^0 | H' | \phi_k^0 \rangle \tag{5.1.1.3}$$

Now, we want to find the first-order approximation of the wave function. Here we examine the case $m \neq k$, here, the first term in the square bracket is zero, and the third term given W_{mk}, thus

$$(E_m^0 - E_k^0)a_m^{(1)} + W_{mk} = 0 \tag{5.1.1.4}$$

given that

$$a_m^{(1)} = \frac{W_{mk}}{E_k^0 - E_m^0} \Rightarrow a_n^{(1)} = \frac{W_{nk}}{E_k^0 - E_n^0} \tag{5.1.1.5}$$

now the expansion parameter is $a_n = a_n^{(0)} + \varepsilon a_n^{(1)}$. Thus, the wave function is

$$\begin{aligned} |\Psi\rangle &= \sum_n a_n |\phi_n^0\rangle = \sum_n (a_n^{(0)} + \varepsilon a_n^{(1)}) |\phi_n^0\rangle \\ &= \sum_n (\delta_{nk} + \varepsilon a_n^{(1)}) |\phi_n^0\rangle \\ &= |\phi_k^0\rangle + \varepsilon \sum_{n(n\neq k)} \frac{W_{nk}}{E_k^0 - E_n^0} |\phi_n^0\rangle \\ &= |\phi_k^0\rangle + \sum_{n(n\neq k)} \frac{H'_{nk}}{E_k^0 - E_n^0} |\phi_n^0\rangle \end{aligned} \tag{5.1.1.6}$$

5.1.2 Second Order Energy Correction

Now, we are going to find the second order correction of the energy, and we take the perturbation formula in second order ε, and take the m index to k.

$$\begin{aligned} &(E_k^0 - E^{(0)})a_k^{(0)} \\ &+ \varepsilon[(W_{kk} - E^{(1)})a_k^{(0)} + (E_k^0 - E^{(0)})a_k^{(1)} + \sum_{n(n\neq k)} W_{kn}a_n^{(0)}] \\ &+ \varepsilon^2[(W_{kk} - E^{(1)})a_k^{(1)} + (E_k^0 - E^{(0)})a_m^{(2)} - E^{(2)}a_k^{(0)} \\ &+ \sum_{n(n\neq k)} W_{kn}a_n^{(1)}] = 0 \end{aligned} \tag{5.1.2.1}$$

take the result for the zero'th order approximation

$$E^{(0)} = E_k^0, \quad a_k^{(0)} = 1 \tag{5.1.2.2}$$

and the first order correction

$$E^{(1)} = W_{kk}, \quad a_n^{(1)} = \frac{W_{nk}}{E_k^0 - E_n^0} \tag{5.1.2.3}$$

now the perturbation formula looks like

$$\begin{aligned} &\varepsilon \sum_{n(n\neq k)} W_{kn}a_n^{(0)} - \varepsilon^2 E^{(2)}a_k^{(0)} + \sum_{n(n\neq k)} W_{kn}a_n^{(1)} \\ =&\varepsilon \sum_{n(n\neq k)} W_{kn}\delta_{nk} - \varepsilon^2 E^{(2)} + \varepsilon^2 \sum_{n(n\neq k)} \frac{W_{kn}W_{nk}}{E_k^0 - E_n^0} \\ =&-\varepsilon^2 E^{(2)} + \sum_{n(n\neq k)} \frac{H'_{kn}H'_{nk}}{E_k^0 - E_n^0} = 0 \end{aligned} \tag{5.1.2.4}$$

here

$$\varepsilon^2 E^{(2)} = -\sum_{n(n\neq k)} \frac{|H'_{nk}|^2}{E_n^0 - E_k^0} \tag{5.1.2.5}$$

this is the second order correction for the system energy, it means that when the system is in the ground state ($k = 1$), we have $E_n^0 > E_k^0$, and $|H'_{nk}|^2$ is positive, thus $\varepsilon^2 E^{(2)} < 0$. This means that the second-order correction of the system energy is negative, no matter the sign of the matrix element H'_{nk}

Similarly, we can also find the higher order correction for the energy and wave function, and the addition of an infinite many order corrections will be the accurate solution. For real-life problems, we take the order as our need for the accuracy, but when perturbation is small enough, it will be accurate enough to take the second-order approximation; thus, the energy and wave function of the k'th order eigenstate are given by

$$\begin{aligned} E &= E_k^0 + H'_{kk} - \sum_{n(n\neq k)} \frac{|H'_{nk}|^2}{E_n^0 - E_k^0} + \cdots \\ |\Psi\rangle &= |\phi_k^0\rangle - \sum_{n(n\neq k)} \frac{H'_{nk}}{E_n^0 - E_k^0} |\phi_n^0\rangle + \cdots \end{aligned} \tag{5.1.2.6}$$

And, an IMPORTANT thing is, the applicable condition for the perturbation theory is that the series in (5.1.2.6) should converge, and to justify the convergence of this series, we should know its general term. However, we don't know the higher-order term. Under this condition, we can only require that the magnitude of the later term should always be less than the prior terms, thus, we have the *Perturbation Theory Applicable Conditions* to be

$$\left| \frac{H'_{nk}}{E_n^0 - E_k^0} \right| \ll 1 \ (n \neq k) \tag{5.1.2.7}$$

it is the case we mentioned at the beginning of this chapter, the H' is very small. Here, we can also notice that we not only require $|H'_{nk}|$ to be small enough, but also require the separation between energy levels $|E_n^0 - E_k^0|$ to be big enough. Thus, for example, in the Hydrogen atom, with the increase of n, the separation of the energy levels becomes smaller and smaller; thus, the applicable condition becomes harder to be satisfy. Thus, in this system, perturbation theory can only be used to find the correction for the lower energy level, instead of the higher energy level.

5.2 Degeneracy Perturbation Theorem

At this moment, we only considered the case where each non-perturbed energy level has only one eigenstate associated. However, in many cases, for example, hydrogen atom, there exist one energy level corresponded with multiple eigenstates, naming N fold degenerate, in this case, we must use *Degenerate Perturbation Theory*

In the degenerated case, the wave function of the system can be written as

$$|\Psi\rangle = \sum_{\beta=1}^{N} \sum_{n} a_{n\beta} |\phi_{n\beta}^0\rangle \tag{5.2.0.1}$$

here, equation (5.1.0.5) should be written in this form

$$(E_m^0 - E + \varepsilon W_{m\alpha m\alpha}) a_{m\alpha} + \varepsilon \sum_{\substack{n\neq m \\ \beta\neq\alpha}} W_{m\alpha n\beta} a_{n\beta} = 0 \tag{5.2.0.2}$$

here, index α and β are equivalent. and E is the system-energy; E_m^0 is the eigenenergy of non-perturbed state, thus it don't have index β; $a_{m\alpha}$ is the expansion parameter; and the matrix element $W_{m\alpha n\beta}$ can be written as

$$W_{m\alpha n\beta} = \langle \phi_{m\alpha}^0 | W | \phi_{n\beta}^0 \rangle \tag{5.2.0.3}$$

First, we consider the zeroth energy approximation under degenerate case, take $m = k$, we have

$$(E_k^0 - E^{(0)}) a_{k\alpha}^{(0)} = 0 \tag{5.2.0.4}$$

Thus, we have

$$E^{(0)} = E_k^0 \quad (a_{k\alpha}^{(0)} \neq 0, \ \alpha = 1, 2, 3, \cdots, N) \tag{5.2.0.5}$$

Thus, under zeroth approximation, the system energy equals to the eigenenergy of a non-perturbed state. We then consider the zeroth approximation of the wave function, here, we first write down the wave function of non-degenerate state

$$|\Psi\rangle = \sum_n a_n |\phi_n^0\rangle = \sum_n a_n^{(0)} |\phi_n^0\rangle = \sum_n \delta_{nk} |\phi_n^0\rangle = |\phi_k^0\rangle \tag{5.2.0.6}$$

where the system wave function is a wave function of a non-degenerate state. Similarly, the zeroth approximation of degenerate state also won't sum over n, but only sum over the degenerated states, thus, it becomes

$$|\Psi\rangle = \sum_{\beta=1}^{N} a_{k\beta} |\phi_{k\beta}^0\rangle \tag{5.2.0.7}$$

Thus, the double summation in (5.2.0.2) becomes a summation only over β under zeroth order approximation. Thus, for eigenstate k, (5.2.0.2) becomes

$$(E_k^0 - E + \varepsilon W_{k\alpha k\alpha}) a_{k\alpha}^{(0)} + \varepsilon \sum_{\beta(\beta\neq\alpha)}^{N} W_{k\alpha k\beta} a_{k\beta}^{(0)} = 0 \tag{5.2.0.8}$$

This is a linear homogeneous system about $a_{k\alpha}^{(0)} (\alpha = 1, 2, 3, \cdots, N)$

$$\begin{bmatrix} E^{(0)} - E + \varepsilon W_{11} & \cdots & \varepsilon W_{1N} \\ \varepsilon W_{21} & \cdots & \varepsilon W_{2N} \\ \vdots & & \vdots \\ \varepsilon W_{N1} & \cdots & E^{(0)} - E + \varepsilon W_{NN} \end{bmatrix} \begin{bmatrix} a_1^{(0)} \\ a_2^{(0)} \\ \vdots \\ a_N^{(0)} \end{bmatrix} = 0 \tag{5.2.0.9}$$

And the sufficient and necessary condition is the coefficient determinant equals to zero

$$\begin{vmatrix} E^{(0)} - E + \varepsilon W_{11} & \cdots & \varepsilon W_{1N} \\ \varepsilon W_{21} & \cdots & \varepsilon W_{2N} \\ \vdots & & \vdots \\ \varepsilon W_{N1} & \cdots & E^{(0)} - E + \varepsilon W_{NN} \end{vmatrix} \tag{5.2.0.10}$$

yields N roots, which serve as the zeroth-order (with first-order ε correction) approximations to the system energy E. If all N energies are distinct, the original N-fold degeneracy is completely lifted at this order. And if some roots remain equal, the degeneracy is only partially removed, and one must proceed to higher-order perturbation theory to fully split the remaining degenerate levels.

Then, using homogeneous linear system to solve the expansion parameters $a_\alpha^{(0)} (\alpha = 1, 2, 3, \cdots, N)$, thus the zeroth order approximation of the wave function can be given by

$$|\Psi\rangle = \sum_{\alpha=1}^{N} a_\alpha^{(0)} |\phi_\alpha^0\rangle \tag{5.2.0.11}$$

5.3 Applications of Steady State Perturbation Theorem

5.3.1 Example 1

6 Spin and Angular Momentum

7 Conclusion

Until this point, you have learned approximately 80% of the foundational concepts in quantum mechanics. The remaining

topics—primarily involving additional operators and more advanced techniques—will likely become familiar as you delve deeper into the subject. With the knowledge gained in this paper, you are well-equipped to explore further complexities in quantum theory, bridging the gap between abstract mathematics and physical applications.

At last, I hope this introduction has not only solidified your understanding of the core principles but also sparked your curiosity to continue your journey into the fascinating world of quantum mechanics!

Appendix Overview

In this paper, we have introduced several key mathematical tools and proofs that support the results presented in the main text. For the convenience of the reader, the following appendices provide detailed derivations and proofs for these foundational concepts:

- **Appendix A: Proof of the Orthogonality of Trigonometric Functions**
- **Appendix B: Exponential Representation of Trigonometric Functions**
- **Appendix C: Fourier Transformation between Position and Momentum Representations**
- **Appendix D: Derivation of the Stationary State Solution**
- **Appendix E: Proof of Common Eigenstates**
- **Appendix F: Proof of the Properties of Hermitian Operators**
- **Appendix G: Proof of Uncertainty Relation**
- **Appendix H: Completeness under General Spectrum**

A Proof of the Orthogonality of Trigonometric Functions

The primary trigonometry function members are

$$1, \cos x, \sin x, \cos 2x, \sin 2x, \cdots \cos nx, \sin nx$$

For any integer $n \neq m$.

$$\int_{-\pi}^{\pi} 1 \cdot \cos nx dx = \int_{-\pi}^{\pi} 1 \cdot \sin nx dx = 0$$

$$\int_{-\pi}^{\pi} \cos mx \cos nx dx = \frac{1}{2}\int_{-\pi}^{\pi} \cos(m-n)x + \cos(m-n)x dx = 0$$

$$\int_{-\pi}^{\pi} \sin mx \sin nx dx = \frac{1}{2}\int_{-\pi}^{\pi} \cos(m-n)x - \cos(m+n)x dx = 0$$

$$\int_{-\pi}^{\pi} \sin mx \cos nx dx = \frac{1}{2}\int_{-\pi}^{\pi} \sin(m+n)x + \sin(m-n)x dx = 0$$

$$\int_{-\pi}^{\pi} \sin nx \cos nx dx = \frac{1}{2}\int_{-\pi}^{\pi} \sin 2nx dx = 0$$

For $n = m$

$$\int_{-\pi}^{\pi} \cos^2 nx dx = \frac{1}{2}\int_{-\pi}^{\pi} 1 + \cos 2nx dx = \pi$$

$$\int_{-\pi}^{\pi} \sin^2 nx dx = \frac{1}{2}\int_{-\pi}^{\pi} 1 - \sin 2nx dx = \pi$$

Thus, by evaluating the inner product of different trigonometry functions, we can conclude that the set of these trigonometry functions form a orthogonal basis, in such case, every periodic functions can be representing by this base.

B The Exponential Representation of Trigonometry functions

The change from trigonometry function to exponential functions can be obtained by the Euler Formula

$$\cos\frac{n\pi}{L}x = \frac{e^{in\pi x/L}+e^{-in\pi x/L}}{2},$$
$$\sin\frac{n\pi}{L}x = \frac{e^{in\pi x/L}-e^{-in\pi x/L}}{2i}.$$

Substitute back into the linear combination of trigonometric functions

$$\begin{aligned} f(x) = a_0 + \sum_{n=1}^{\infty}\Bigg[&\frac{a_n}{2}\left(e^{in\pi x/L}+e^{-in\pi x/L}\right) \\ &-\frac{ib_n}{2}\left(e^{in\pi x/L}-e^{-in\pi x/L}\right)\Bigg] \end{aligned} \tag{B.1}$$

Rearrange

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left[\frac{a_n-ib_n}{2}e^{in\pi x/L}+\frac{a_n+ib_n}{2}e^{-in\pi x/L}\right]$$

and if we allow n to be negative integers

$$a_{-n} = \frac{1}{L}\int_{-L}^{L}\cos\frac{-n\pi x}{L}f(x)dx = a_n$$

$$b_{-n} = \frac{1}{L}\int_{-L}^{L}\sin\frac{-n\pi x}{L}f(x)dx = -b_n$$

And let $c_0 = \frac{a_0}{2}$, $c_n = \frac{a_n-ib_n}{2}$, $c_n^* = \frac{a_n-ib_n}{2}$,

$$\begin{aligned} f(x) =& a_0 + \sum_{n=1}^{\infty}\frac{a_n-ib_n}{2}e^{in\pi x/L} + \sum_{n=1}^{\infty}\frac{a_{-n}-ib_{-n}}{2}e^{-in\pi x/L} \\ =& a_0 + \sum_{n=1}^{\infty}\frac{a_n-ib_n}{2}e^{in\pi x/L} + \sum_{n=-1}^{-\infty}\frac{a_n-ib_n}{2}e^{-in\pi x/L} \\ =& c_0 + \sum_{i=1}^{\infty}c_n e^{in\pi x/L} + \sum_{i=-1}^{-\infty}c_n e^{in\pi x/L} \\ f(x) =& c_0 + \sum_{n=-\infty}^{\infty}c_n e^{in\pi x/L} \end{aligned}$$

With the coefficient

$$c_n = \frac{1}{2L}\left[\int_{-L}^{L}e^{-in\pi x/L}f(x)\,dx\right]$$

And the final exponential representation is given by

$$\begin{aligned} f(x) &= \sum_{n=-\infty}^{\infty}c_n e^{in\pi x/L} \\ c_n &= \frac{1}{2L}\int_{-L}^{L}e^{-in\pi x/L}f(x)\,dx \end{aligned} \tag{B.2}$$

C The Fourier Transformation between Position and Momentum Representation

As the exponential representation is given by(B.2), with the relationship

$$p_n = \frac{h}{\lambda} = \hbar k_n = \frac{\hbar\pi n}{L}$$

When $L\to\infty$, $dp_n = \hbar\frac{\pi}{L}dn$ we obtain

$$\begin{aligned} f(x) &= \sum_{n=-\infty}^{\infty}c_n e^{in\pi x/L} \\ &= \int_{-\infty}^{\infty}e^{ip_n x/\hbar}\frac{2Lc_n}{2\pi\hbar}\,dp \\ &= \frac{1}{\sqrt{h}}\int_{-\infty}^{\infty}e^{ipx/\hbar}\frac{2Lc_n}{\sqrt{h}}\,dp \end{aligned}$$

Take $C(p) = \frac{2L}{\sqrt{h}}c_n$

$$\begin{aligned} f(x) &= \int_{-\infty}^{\infty}e^{ipx/\hbar}C(p)\,dp \\ C(p) &= \frac{2L}{\sqrt{\hbar}}\frac{1}{2L}\int_{-\infty}^{\infty}e^{-ipx/\hbar}f(x)\,dx \\ &= \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty}e^{-ipx/\hbar}f(x)\,dx \end{aligned}$$

Finally,

$$\begin{aligned} f(x,t) = \psi(x,t) &= \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty}e^{ipx/\hbar}C(p)\,dp \\ C(p,t) = \psi(p,t) &= \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty}e^{-ipx/\hbar}f(x)\,dx \end{aligned} \tag{C.1}$$

D Derivation of Stationary State Solution

The stationary state solution can be obtained by separation of variables.

$$\Psi(x,t) = \psi(x)f(t)$$

Substitute back into the equation

$$\frac{i\hbar}{f(t)}\frac{df(t)}{dt} = \frac{1}{\psi(x)}\left[-\frac{\hbar^2}{2m}\nabla^2\psi(x)+V(x)\psi(x)\right]$$

Since the variables between two sides of this equation is different, we can let the two side both equal to a constant.

$$\begin{aligned} i\hbar\frac{df}{dt} &= Ef \\ -\frac{\hbar^2}{2m}\nabla^2\psi + V(x)\psi &= E\psi \end{aligned}$$

The solution of first equation is

$$f = Ce^{-iEt/\hbar}$$

Thus a form of solution for stationary state Schrödinger's Equation is $\psi(x,t) = \psi_E(x)e^{-iEt/\hbar}$

E Proof of Common Eigenstates:

Suppose ψ is an eigenstate of A with eigenvalue a, so that

$$A\psi = a\psi.$$

If A and B commute, we have

$$[A,B]\psi = (AB-BA)\psi = 0.$$

This implies that

$$AB\psi = BA\psi.$$

Substituting $A\psi = a\psi$ into the right-hand side yields

$$AB\psi = B(a\psi) = a(B\psi).$$

Thus, $B\psi$ is also an eigenstate of A with the same eigenvalue a. If the eigenvalue a is non-degenerate, it follows that

$$B\psi = b\psi,$$

for some scalar b. Therefore, ψ is a common eigenstate of both A and B. In the case of degeneracy, one can always choose a basis within the degenerate subspace such that all basis vectors are eigenstates of both A and B.

F Proof for the Properties of Hermitian

4. **Hermitian operators have real eigenvalues.**

   Suppose $\hat{A}$ is a Hermitian operator and ψ is an eigenfunction of $\hat{A}$ with eigenvalue λ, i.e.,

   $$\hat{A}\psi = \lambda\psi.$$

   Taking the inner product with ψ yields

   $$\langle\psi, \hat{A}\psi\rangle = \lambda\langle\psi, \psi\rangle.$$

   Since $\hat{A}$ is Hermitian, we have

   $$\langle\psi, \hat{A}\psi\rangle = \langle\hat{A}\psi, \psi\rangle^*,$$

   which implies that $\langle\psi, \hat{A}\psi\rangle$ is real. Because $\langle\psi, \psi\rangle > 0$ for a nonzero ψ, it follows that λ must be real.

5. **Eigenstates corresponding to different eigenvalues are orthogonal.**

   Let ψ_1 and ψ_2 be eigenfunctions of the Hermitian operator $\hat{A}$ corresponding to eigenvalues λ_1 and λ_2 respectively, with $\lambda_1 \neq \lambda_2$:

   $$\hat{A}\psi_1 = \lambda_1\psi_1, \quad \hat{A}\psi_2 = \lambda_2\psi_2.$$

   Taking the inner product of the first equation with ψ_2 gives:

   $$\langle\psi_2, \hat{A}\psi_1\rangle = \lambda_1\langle\psi_2, \psi_1\rangle.$$

   Since $\hat{A}$ is Hermitian,

   $$\langle\psi_2, \hat{A}\psi_1\rangle = \langle\hat{A}\psi_2, \psi_1\rangle^* = \lambda_2\langle\psi_2, \psi_1\rangle.$$

   Therefore,

   $$\lambda_1\langle\psi_2, \psi_1\rangle = \lambda_2\langle\psi_2, \psi_1\rangle.$$

   Since $\lambda_1 \neq \lambda_2$, it must be that

   $$\langle\psi_2, \psi_1\rangle = 0,$$

   proving that ψ_1 and ψ_2 are orthogonal.

6. **Eigenstates corresponding to the same eigenvalue can be chosen to be orthogonal.**

   In the case of degenerate eigenvalues (i.e., when an eigenvalue λ has more than one linearly independent eigenfunction), the corresponding eigenspace is a subspace of the Hilbert space. In any inner product space, any set of linearly independent vectors can be orthogonalized using the Gram–Schmidt process. Therefore, within the degenerate subspace, one can always choose an orthonormal basis such that the eigenstates are mutually orthogonal.

G Proof of Uncertainty Relation

As we introduced in section 3.3.1, the variance of observable is

$$\overline{(\Delta\hat{A})^2} = \langle A^2\rangle - \langle A\rangle^2, \quad \overline{(\Delta\hat{B})^2} = \langle B^2\rangle - \langle B\rangle^2$$

Consider a uon-negative average of any state

$$I(\xi) = \int |(\hat{A} + i\xi\hat{B})\psi|^2 \, d\tau \geq 0$$

where ξ is real-valued coefficient.

$$\begin{aligned} I(\xi) =& \int [(\hat{A} + i\xi\hat{B})\psi]^*(\hat{A} + i\xi\hat{B})\psi \, d\tau \\ =& \int [(\hat{A}\psi)^*\hat{A}\psi + i\xi(\hat{A}\psi)^*\hat{B}\psi - \\ & i\xi(\hat{B}\psi)^*\hat{A}\psi + (\hat{B}\psi)^*\hat{B}\psi]d\tau \\ =& \int (\psi^*\hat{A}^2\psi + i\xi\psi^*\hat{A}\hat{B}\psi - i\xi\psi^*\hat{B}\hat{A}\psi + \psi^*\hat{B}^2\psi) \, d\tau \\ =& \overline{A^2} + i\xi\overline{[\hat{A}, \hat{B}]} + \xi^2\overline{B^2} \geq 0 \end{aligned}$$

Thus, for this inequality to be satisfy for any ξ, should require

$$\{i\overline{[\hat{A}, \hat{B}]}\}^2 - 4\overline{A^2} \cdot \overline{B^2} \leq 0$$

or

$$\overline{A^2} \cdot \overline{B^2} \geq \frac{1}{4}(i\overline{[\hat{A}, \hat{B}]})^2$$

And if we use $\Delta\hat{A} = \hat{A} - \overline{A}$, $\Delta\hat{B} = \hat{B} - \overline{B}$ to substitute $\hat{A}, \hat{B}$, we have

$$[\Delta\hat{A}, \Delta\hat{B}] = [\hat{A} - \overline{A}, \hat{B} - \overline{B}] = [\hat{A}, \hat{B}]$$

Then we get

$$\overline{(\Delta A)^2} \cdot \overline{(\Delta B)^2} \geq \frac{1}{4}(i\overline{[\hat{A}, \hat{B}]})^2 \tag{G.1}$$

Which is the general representation of uncertainty principle. Specially, if $\hat{A}, \hat{B}$ are conjugate observables, they have commute to be purely complex number. And a significant example is position and momentum, which

$$[\hat{x}, \hat{p}_x] = i\hbar$$

$$\overline{(\Delta x)^2} \cdot \overline{(\Delta p_x)^2} \geq \frac{\hbar^2}{4}$$

For simplify, we can take the square root for both side

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Which is the most familiar format of the uncertainty relation.

H Completeness under General Spectrum

We already discussed the case where we have discrete spectrum, in many cases, the spectrum is not continuous or either continuous and discrete (General Form), in fact, in these cases, the completeness still satisfies.

H.1 Analysis of Continuous Spectrum

Suppose $\hat{G}$ have continuous eigenvalue spectrum, for eigenvalue λ, the eigen-wave-function is φ_λ.

$$\hat{G}\varphi_\lambda = \lambda\varphi_\lambda$$

Then, for any wave function φ, can be expanded with $\{\varphi_\lambda\}$,

$$\varphi = \int C_\lambda \varphi_\lambda \, d\lambda$$

If φ has already normalized, then expansion coefficients can be determined by

$$C_\lambda = \int \varphi_\lambda^* \varphi \, d\tau \tag{H.1}$$

where we say $|C_\lambda|^2 d\lambda$ is the probability for observable $\hat{G}$ to exist between λ and $\lambda + d\lambda$.
To have (H.1), must have requirement for the orthonormal condition of eigen-wave-function, since

$$\begin{aligned} C_\lambda &= \int \varphi_\lambda^* \varphi \, d\tau = \int \varphi_\lambda^* \left(\int C_\lambda \varphi_\lambda \, d\lambda \right) d\tau \\ &= \int C'_\lambda \left(\int \varphi_\lambda \varphi_{\lambda'}, d\tau \right) d\lambda' \end{aligned}$$

where φ are arbitrary, and thus C_λ are arbitrary, to satisfy the equation above, meaning that

$$\int \varphi_\lambda^* \varphi_{\lambda'} \, d\tau = \delta(\lambda - \lambda') \tag{H.2}$$

Equation (H.2) is the normalization condition requirement for a continuous spectrum to be satisfied.
Then, we solved the problem of normalize eigen-wave-function in continuous spectrum, we can see, a better normalization idea is not normalize to 1, but to δ function.
For example, the eigen-wave-function of momentum,

$$\varphi_p = Ce^{ipx/\hbar}$$

using the orthonormal relation (H.2), we have

$$\begin{aligned} \delta(p' - p) =& \int \varphi_p^* \varphi_p \, dx = \int C^* C e^{-i(p'-p)x/\hbar} dx \\ =& 2\pi\hbar\delta C^* C(p' - p) \\ C^* C =& \frac{1}{2\pi\hbar} \\ C =& \frac{1}{\sqrt{2\pi\hbar}} \end{aligned}$$

In three dimensional case

$$\begin{aligned} \varphi_{\mathbf{p}}(\mathbf{x}) =& Ce^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \\ \int \varphi_{\mathbf{p}'}^*(\mathbf{x}) \varphi_{\mathbf{p}}(\mathbf{x}) \, d\tau =& \delta^{(3)}(\mathbf{p}' - \mathbf{p}) \\ C =& \frac{1}{(2\pi\hbar)^{3/2}} \end{aligned}$$

Thus, the average of $\hat{G}$ in continuous spectrum is

$$\begin{aligned} \overline{G} &= \int \varphi^* \hat{G} \varphi \, d\tau = \int \left(\int C_\lambda^* \varphi_\lambda^* \, d\lambda \right) \hat{G} \left(\int C_{\lambda'} \varphi_{\lambda'} \, d\lambda' \right) d\tau \\ &= \int C_\lambda^* C_{\lambda'} \lambda' d\lambda' d\lambda \int \varphi_\lambda^* \varphi_{\lambda'} d\tau = \int \lambda C_\lambda^* C_\lambda d\lambda \end{aligned} \tag{H.3}$$

where it is clearly to see C^*C is satisfied the requirement to be the distribution density.

H.2 Analysis of General Spectrum

Generally speaking, an observable can both have discrete eigenvalue and continuous eigenvalue.
These eigen-wave-function satisfies the orthonormal condition

$$\begin{cases} \int \varphi_m^* \varphi_n \, d\tau = \delta_{mn} \\ \int \varphi_\lambda^* \varphi_{\lambda'} \, d\tau = \delta(\lambda - \lambda') \\ \int \varphi_n^* \varphi_\lambda \, d\tau = 0 \end{cases} \tag{H.4}$$

All these eigen-wave-functions are orthonormal with each other, thus, it forms a complete orthonormal coordinate, and wave function can be expanded with it.

$$\begin{aligned} \varphi =& \sum_n C_n \varphi_n + \int C_\lambda \varphi_\lambda d\lambda \\ C_n =& \int \varphi_n^* \varphi \, d\tau \\ C_\lambda =& \int \varphi_\lambda^* \varphi \, d\tau \end{aligned} \tag{H.5}$$

with normalized φ have

$$\sum |C_n|^2 + \int |C_\lambda|^2 d\lambda = 1 \tag{H.6}$$

in which $|C_n|^2$ represent the probability $\hat{F}$ have value f_n, and $|C_\lambda|^2$ represent the probability $\hat{F}$ have value in between λ and $\lambda + d\lambda$. And the average

$$\overline{F} = \int \varphi^* \hat{F} \varphi \, d\tau = \sum |C_n|^2 f_n + \int |C_\lambda|^2 d\lambda$$

here we already know how to find the distribution for an observable having general spectrum, these rely on the completeness of eigenstate and this can be represented by another form:

H.3 Completeness relation

By (H.6), we have

$$\begin{aligned} \varphi(\mathbf{x}) =& \sum C_n \varphi_n(\mathbf{x}) + \int C_\lambda \varphi_\lambda(\mathbf{x}) d\lambda \\ =& \sum_n \left[\int \varphi_n^*(\mathbf{x}') \varphi(\mathbf{x}') d\tau' \right] \varphi_n(\mathbf{x}) \\ &+ \int \left[\int \varphi_n^*(\mathbf{x}') \varphi(\mathbf{x}') d\tau' \right] \varphi_n(\mathbf{x}) \, d\lambda \\ =& \int \varphi(\mathbf{x}') \left[\sum_n \varphi_n^*(\mathbf{x}') \varphi(\mathbf{x}) + \int \varphi_\lambda^*(\mathbf{x}') \varphi_\lambda(\mathbf{x}) \, d\lambda \right] d\tau' \end{aligned} \tag{H.7}$$

Since φ is arbitrary, by property of δ function

$$\sum_n \varphi_n^*(\mathbf{x}') \varphi(\mathbf{x}) + \int \varphi_\lambda^*(\mathbf{x}') \varphi_\lambda(\mathbf{x}) \, d\lambda = \delta^{(3)}(\mathbf{x}' - \mathbf{x}) \tag{H.8}$$

this is called the condition for completeness, one can also called the enclosure of wave function.
Thus, as you can see, it is easy to expand our condition from discrete spectrum to general spectrum, by simplicity, for general discussion, we only deal with discrete spectrum, and the procedure above can be a reference on the way we generalize our discussion to general spectrum.

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